MILITARY SPECIFICATION

MODULES, STANDARD ELECTRONIC

MEMORY, READ ONLY, DIGITAL

This specification is approved for use by the Naval Electronic Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense

1. SCOPE

1.1 Scope. This document establishes the requirements and test procedures for the acquisition of an electronic digital read only memory module.

1.2 Classification. This specification covers two classes of modules.

Class I and class II as specified in MIL-STD-1389.

2. APPLICABLE DOCUMENTS

2.1 Government specifications and standards. Unless otherwise specified, the following specifications and standards, of the issue listed in that issue of the Department of Defense Index of Specifications and Standards specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-M-28787 - Modules, Standard Electronic, General Specification for.

STANDARDS

MILITARY

MIL-STD-1389 - Design Requirements for Standard Electronic Modules.

MIL-STD-1665 - Test Equipment for the Standard Electronic Modules Program.

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 General requirements. The module shall comply with the requirements of MIL-M-28787 except as modified herein.

3.2 Correlation requirements. Before any production modules are tested, all testers and test equipment shall be correlated in accordance with MIL-M-28787.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Naval Electronic Systems Command, ATTN: ELEX 8111, Department of the Navy, Washington, DC 20363, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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3.3 Failure rate requirements. All modules acquired to this specification shall have a predicted failure rate no greater than 0.1031 failures per million operating hours for class I or 0.4500 failures per million operating hours for class II, when predicted in accordance with MIL-STD-1389. When a class II module is used in a class I environment, failure rate for class I shall be used for reliability prediction (see 6.3).

3.4 Performance requirements. The module shall conform to the electrical requirements specified in table I. Class I modules shall conform to the electrical requirements of class I only, and class II modules shall conform to the electrical requirements of class I and class II (see 6.3).

3.5 Module configuration requirements. The module configuration requirements shall be in accordance with MIL-STD-1389 and as specified herein.

3.5.1 Key code. The module key code shall be as specified in the user's detailed module document (6.5.3).

3.5.2 Size. The module size shall be 1A.

3.5.3 Weight. The module shall have a maximum weight of 35 grams.

3.5.4 Contact pins. The module shall have 40 contact pins positioned A1 through A40.

3.5.5 Fin position. The module shall have an insertion and extraction and heat dissipation fin on increment A.

3.5.6 Depth. The module shall have no additional depth requirements.

3.5.7 Holddown holes. The module shall have no additional requirements for location or quantity of holddown holes.

* 3.5.8 Marking. Marking shall be as specified in MIL-STD-1389, and the module shall be marked with the military part number M28787/316-1 for class I and M28787/316-2 for class II and as specified in the user's detailed module document, except no marking shall interfere with heat dissipation from the fin (for example, affixing a label which may be a heat barrier).

3.6 Circuit pin configuration. The function diagram and pin assignments shall conform to figure 1.

3.7 Acceptance tests. All production modules shall be tested in accordance with footnotes 1 and 2 of table I.

3.8 End-of-life limits. Modules which have been plugged into an operating system, or exceed 2 years from date of manufacture and are to be tested, shall be capable of conforming to the 25°C end-of-life limits of table I (see 6.2).

4. QUALITY ASSURANCE PROVISIONS

4.1 Quality assurance. The quality assurance provisions of MIL-M-28787 shall apply.

4.2 Acceptance tests. Acceptance tests of all modules shall be in accordance with procedures and equipment specified in MIL-M-28787.

4.3 Test equipment. Test equipment specified on figures and in tables with a type designator shall be as specified in MIL-STD-1665. Other equipment and components shall be as specified on the figures. All resistors shall be metal film and all capacitors shall be ceramic unless otherwise specified.

4.4 Test procedures. The module shall be tested as specified in table I.

4.4.1 Alternate test methods. Alternate test methods may be employed if approved by the Standard Electronic Modules Program Quality Assurance Activity and they conform to the requirement of MIL-M-28787.

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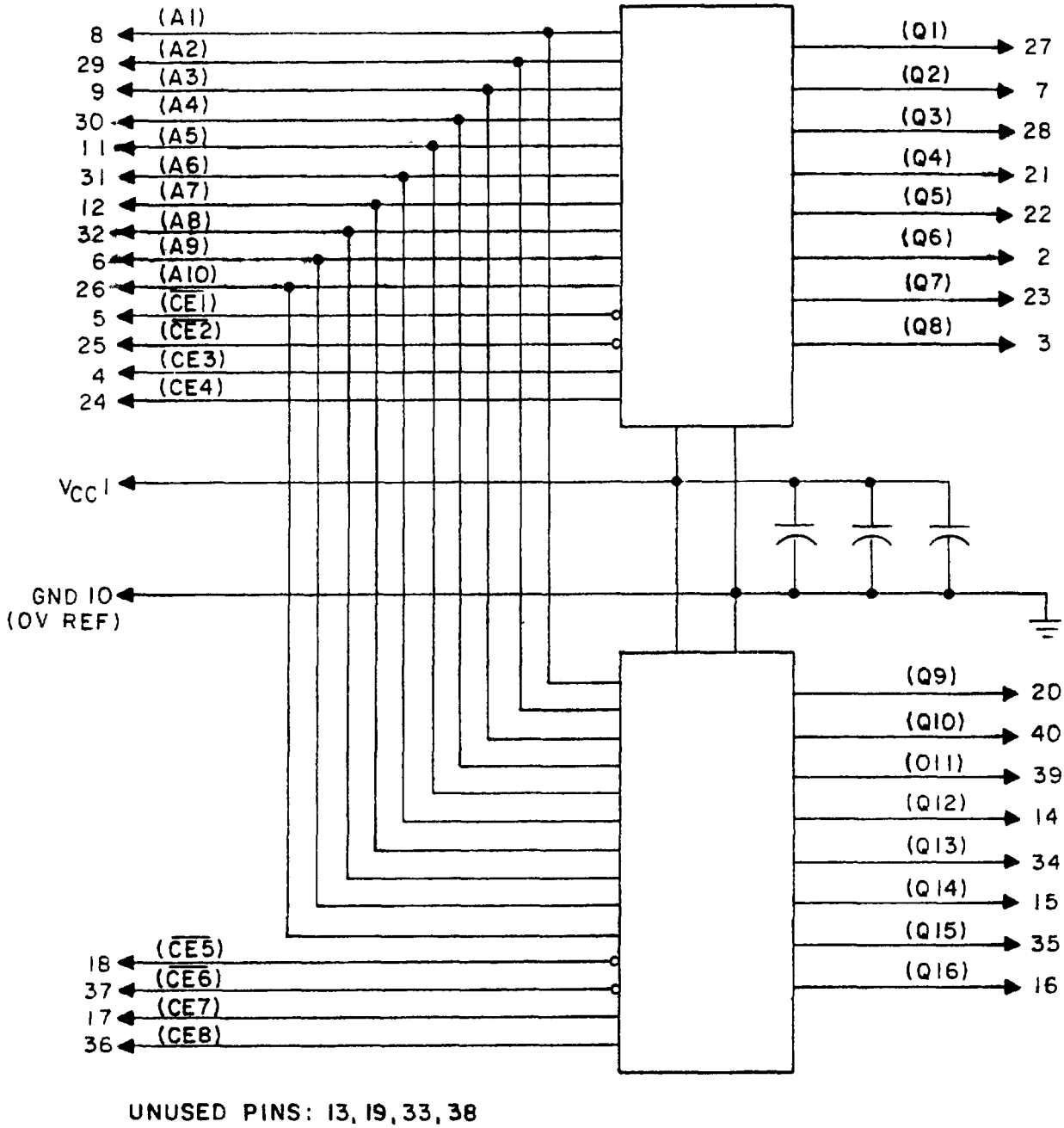


FIGURE 1. Function diagram, read only memory.

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TABLE I. Module electrical requirements.

Test condition	Parameter	Ref test table	Test type 1/-3/	25°C ±5°C 4/		Class I 4/				Class II 4/				Unit
						0°C ±5°C		60°C ±5°C		-55°C ±5°C		85°C ±5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
A1	Isolation													
	Initial	III	S	10	∞	10	∞	10	∞	10	∞	10	∞	MΩ
	End-of-life	III	S	1	∞	1	∞	1	∞	1	∞	1	∞	MΩ
A2-A5	Isolation, used to unused pins													
	Initial	III	S	10	∞	10	∞	10	∞	10	∞	10	∞	MΩ
	End-of-life	III	S	1	∞	1	∞	1	∞	1	∞	1	∞	MΩ
B	V_{CC} stress test	IV	S	← Must pass remaining tests →										
C1-C10	Input breakdown current	V	S	0	2	0	2	0	2	0	2	0	2	mA
C11-C18	Input breakdown current	V	S	0	1	0	1	0	1	0	1	0	1	mA
D1-D16	Output short circuit current	VI	X	-15	-100	-15	-100	-15	-100	-15	-100	-15	-100	mA
E1-E16	Output leakage current, low level	VII	X	-100	100	-100	100	-100	100	-100	100	-100	100	µA
F1-F16	Output leakage current, high level	VIII	X	-100	100	-100	100	-100	100	-100	100	-100	100	µA
G1-G18	Input clamp diode voltage	IX	X	-0.4	-1.2	-0.4	-1.2	-0.4	-1.2	-0.4	-1.2	-0.4	-1.2	V
H1-H16	Output voltage, low level	X	X	0.0	0.5	0.0	0.5	0.0	0.5	0.0	0.5	0.0	0.5	V
J1-J16	Output voltage, high level	XI	X	2.4	4.5	2.4	4.5	2.4	4.5	2.4	4.5	2.4	4.5	V
K1-K10	Input current, high level	XII	X	0	100	0	100	0	100	0	100	0	100	µA
K11-K18	Input current, high level	XII	X	0	50	0	50	0	50	0	50	0	50	µA
L1-L10	Input current, low level	XIII	X	0	-550	0	-550	0	-550	0	-550	0	-550	µA
L11-L18	Input current, low level	XIII	X	0	-275	0	-275	0	-275	0	-275	0	-275	µA
M	Power supply current	XIV	X	10	370	10	370	10	370	10	370	10	370	mA

See footnotes at end of table.

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* TABLE I. Module electrical requirements - Continued.

Test condition	Parameter	Ref test table	Test type 1/-3/	25°C ±5°C 4/		Class I 4/				Class II 4/				Unit
						0°C ±5°C		60°C ±5°C		-55°C ±5°C		85°C ±5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
N1-N32	Enable recovery time (t_{ERL}), C_L ≤10 pF	XV	S	1	50	1	50	1	50	1	50	1	50	ns
N33-N64	Enable recovery time (t_{ERH}), C_L ≤10 pF	XV	S	1	50	1	50	1	50	1	50	1	50	ns
P1, P2	Dynamic	XVI	X	← Must pass →										
P3, P4	Dynamic	XVI	Q	← Must pass →										
R1-R8	Input crosstalk, low level	XVII	Q	0	400	0	400	0	400	0	400	0	400	mV
R9-R16	Input crosstalk, high level	XVII	Q	0	400	0	400	0	400	0	400	0	400	mV
S	Functional	XVIII	X	← Must pass →										
T	Power supply transient amplitude	XIX	Q	0	400	0	400	0	400	0	400	0	400	mV
U	Power supply filter capacitance	XX	S	1.6	3.0	1.6	3.0	1.6	3.0	1.6	3.0	1.6	3.0	µF
V1-V10	Input capacitance	XXI	Q	2	30	2	30	2	30	2	30	2	30	pF
V11-V18	Input capacitance	XXI	Q	2	15	2	15	2	15	2	15	2	15	pF
W1-W16	Output capacitance	XXII	Q	2	15	2	15	2	15	2	15	2	15	pF

1/ "X" parameters shall be 100 percent tested during module acceptance tests.

2/ "S" parameters shall be sample acceptance tested (1 percent or two modules from each inspection lot, whichever is greater) at 25°C.

3/ "Q" parameters shall be tested during qualification testing only.

4/ Unless otherwise specified, limits are for initial and end-of-life.

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4.4.2 Order of testing. Test conditions A through W shall be performed in any order with the following exceptions. If test conditions A, B, C, or D are part of a sequence of tests, the order of testing shall be as follows: A, B, C, D where applicable, followed by the remaining tests in the sequence.

4.5 Qualification testing. Initial qualification tests and periodic-check tests shall be performed in accordance with the test plan specified in MIL-M-28787 with the following modifications or additions.

* 4.5.1 Operating temperature tests. The operating temperature tests shall be conducted in accordance with MIL-M-28787. Before starting these tests, test conditions A through W in table I shall be performed in the order specified in 4.4.2 using the initial limits at 25°C ±5°C ambient. When electrical testing is called for at the end of each temperature stabilization period, test conditions D through U shall be performed in the order specified in 4.4.2 using the initial limits for applicable operating temperature. At all other times during the operating temperature tests, the module shall be operated in the test circuit shown on figure 3.

4.5.1.1 Component temperature test. The component temperature test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 5.

* 4.5.2 Vibration tests. The vibration tests shall be conducted in accordance with MIL-M-28787. During the vibration tests, the module shall be operated in the test circuit shown on figure 6. Following the vibration tests, the electrical tests of table I which are marked with an "X" in the test type column and test condition U shall be performed using the initial limits at 25°C ±5°C ambient.

* 4.5.3 Life test. The life test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 3. Following the life test and while still at the maximum class temperature, electrical tests D through U shall be performed in the order specified in 4.4.2 using the operating temperature end-of-life requirements of table I. Following the return of the module to 25°C, electrical tests D through U shall be performed in the order specified in 4.4.2 using the 25°C ±5°C ambient end-of-life requirements of table I.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-28787.

6. NOTES

6.1 Definitions. Definitions are as specified in MIL-STD-1331.

6.2 End-of-life. When any individual module circuit parameter fails to conform to the end-of-life requirements of table I, the module is considered to have completed its useful life (see 3.8).

6.3 Substitutability. Class II modules are substitutable for class I modules. When class II modules are used in a class I environment, the failure rate for class I shall be used for reliability prediction (see 3.3 and 3.4).

6.4 Module description. The following electrical parameters reflect the maximum class temperature end-of-life limits from table I.

	Class I	Class II	Unit
Maximum power dissipation (outputs not loaded)	2.04	2.04	W
Input voltage			
Low level	-0.5 min, 0.8 max	-0.5 min, 0.8 max	V
High level	2 min, V_{CC} max	2 min, V_{CC} max	V
Input current			
Address			
Low level (V_{IL} = 0.45 V)	-500 max	-500 max	µA
Low level (V_{IL} = 0 V)	-550 max	-550 max	µA
High level (V_{IH} = 2.4 V)	100 max	100 max	µA
High level (V_{IH} = 5.5 V)	2 max	2 max	mA

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	Class I	Class II	Unit
Input current			
Enable			
Low level (V_{IL} = 0.45 V)	-250 max	-250 max	µA
Low level (V_{IL} = 0 V)	-275 max	-275 max	µA
High level (V_{IH} = 2.4 V)	50 max	50 max	µA
High level (V_{IH} = 5.5 V)	1 max	1 max	mA
Output drive current maximum			
Low level (V_{OL} = 0.5 V)	16 max	16 max	mA
High level (V_{OH} = 2.4 V)	-2 max	-2 max	mA

6.5 Module function. The read only memory module is organized as two nonindependent 1024-word by 8-bit memories. The module can be configured as 1024k x 16 bits or 2048k x 8 bits.

* 6.5.1 Input to output relationship. The module contains two nonindependent 1024-word by 8-bit read only memories (see figure 1). When enables $\overline{CE1}$, $\overline{CE5}$, and $\overline{CE2}$, $\overline{CE6}$ are low and enables CE3, CE7 and CE4, CE8 are high, the data programmed at the selected address input appears at the data outputs of that memory. All other enable combinations disable the chip to a high impedance state. The dynamic characteristic waveforms and module truth table are shown on figure 2 and in table II, respectively.

6.5.2 Paralleling inputs and outputs. Corresponding address inputs of two address groups are paralleled for nonindependent operation of the two memory sections. A number of memory outputs can be connected to a common bus. All of the memories except one are placed in the high impedance state and the selected memory is enabled. When outputs of several modules are bussed together, all address lines should be addressed using synchronous clocking.

6.5.3 Application requirements. This specification establishes the general requirements and test procedures for a read only memory (ROM) module. When a SEMP user determines a requirement for a ROM, the user shall prepare a design approval request (DAR) and submit the DAR to the SEMP Design Review Activity (SEMP-DRA) in accordance with MIL-STD-1378. The module user shall prepare a separate module specification which specifies, as a minimum, the module key code as assigned by the SEMP-DRA, the firmware pattern, detailed module marking, and requirements specified herein.

6.6 Changes from previous issue. The margins of this specification are marked with asterisks to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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TABLE II. Module truth table.

CHIP ENABLES	OUTPUTS	
$\overline{CE1}$ = H or $\overline{CE2}$ = H or CE3 = L or CE4 = L $\overline{CE1}$ = L and $\overline{CE2}$ = L and CE3 = H and CE4 = H	High Z Read data of A1-A10	Q1-Q8
$\overline{CE5}$ = H or $\overline{CE6}$ = H or CE7 = L or CE8 = L $\overline{CE5}$ = L and $\overline{CE6}$ = L and CE7 = H and CE8 = H	High Z Read data of A1-A10	Q9-Q16

H = high level input
L = low level input
High Z = high impedance state

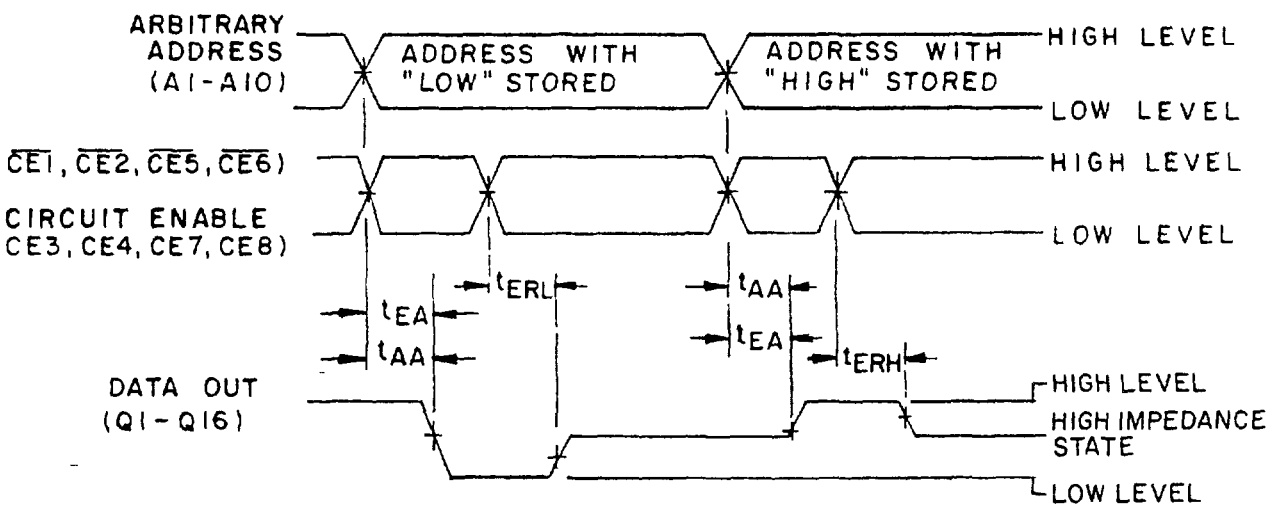
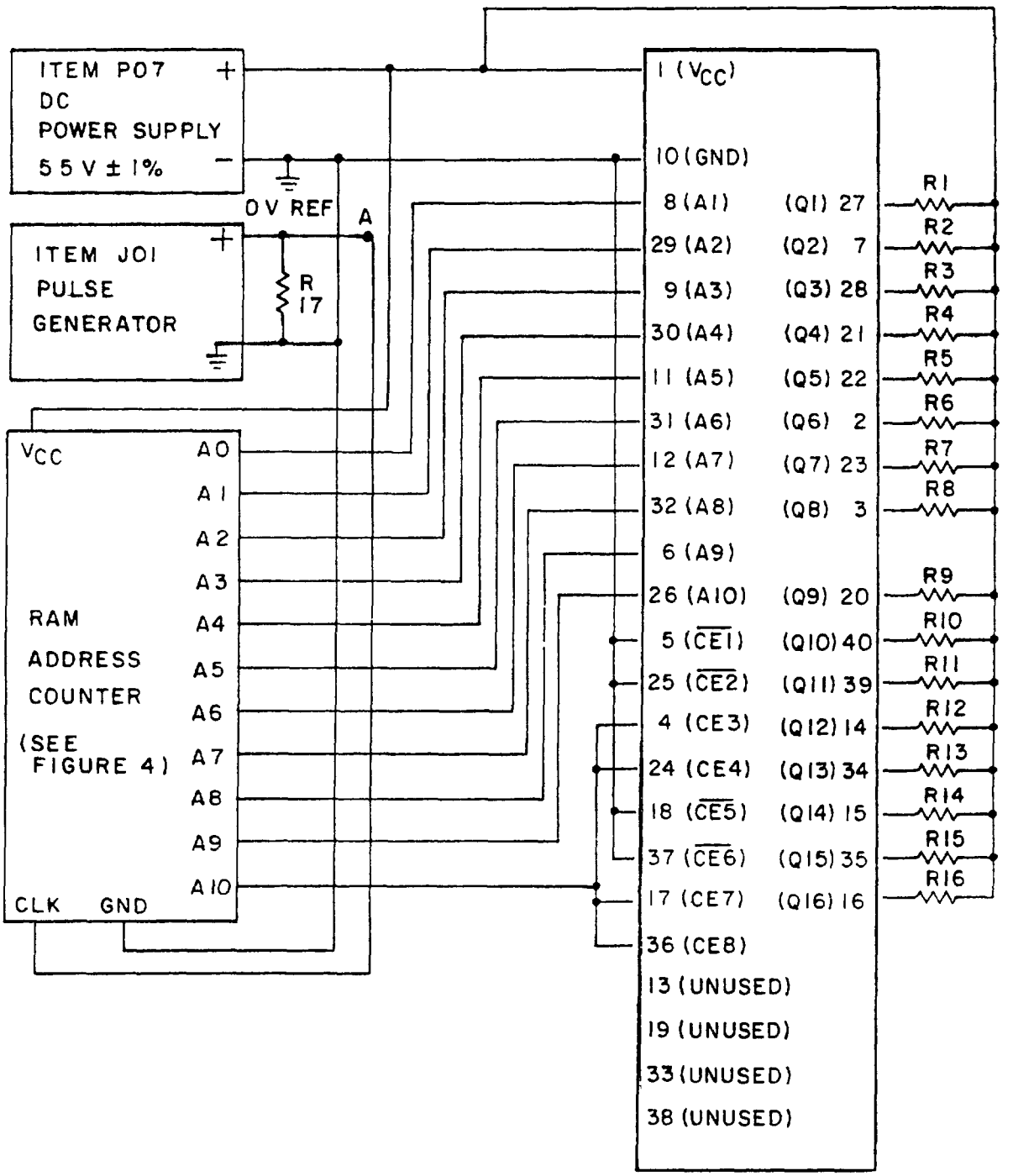


FIGURE 2. Dynamic characteristics waveforms.

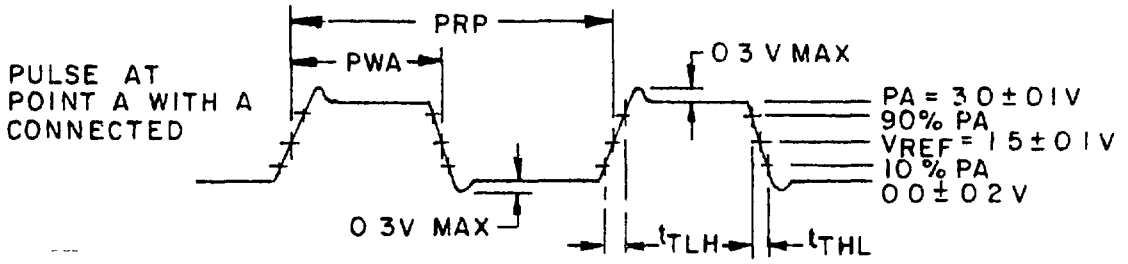
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NOTES:
1. R1 - R16 = 523Ω ±1%, 1/8 W.
2. R17 = 51Ω ±5%, 1 W.

FIGURE 3. Operating temperature and life test circuit.

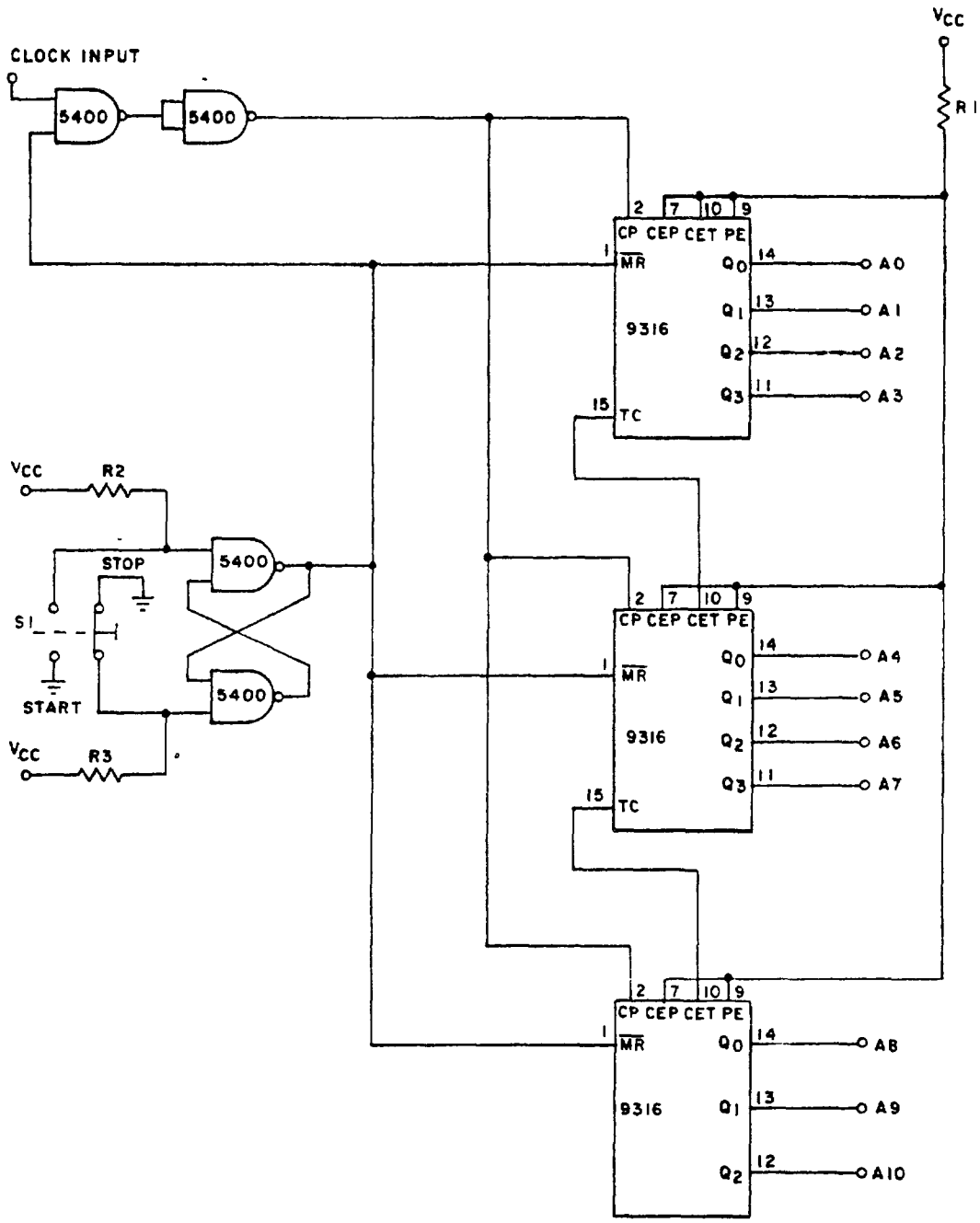
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NOTE: Input pulse parameters:
$t_{TLH} = t_{THL} = 20 \pm 10$ ns
PWA = 100 ±10 ns.
PRP = 200 ±10 ns

FIGURE 3. Operating temperature and life test circuit - Continued.

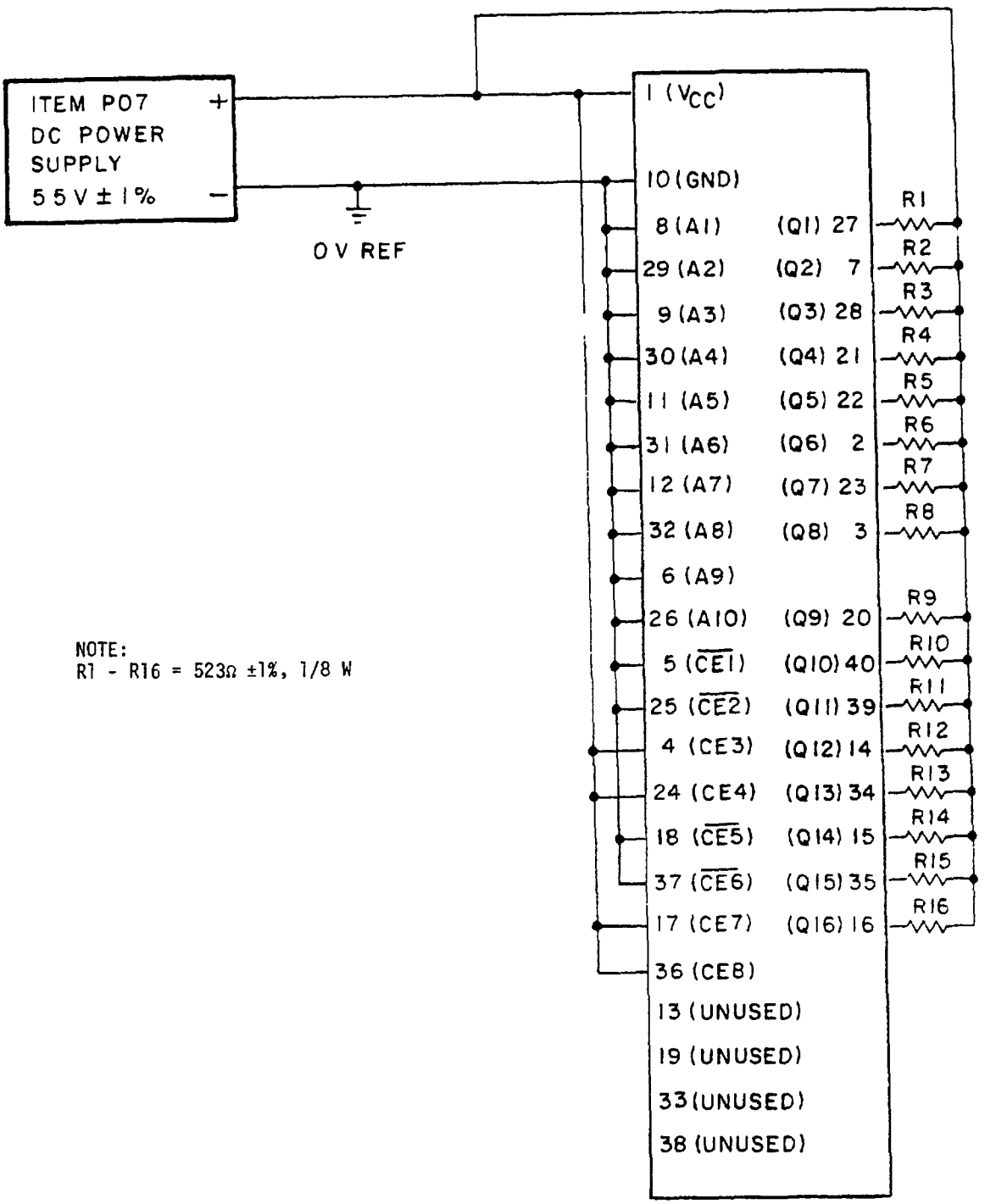
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NOTES:
1. R1 - R3 = 2.2 kΩ ±10%, 1/8 W.
2. S1 = Push button switch (push on, push off).
3. 3 each 9316 high-speed synchronous 4-bit binary counters.
4. 1 each 5400 quad 2-input NAND gate.

FIGURE 4. RAM address counter circuit.

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NOTE:
R1 - R16 = 523Ω ±1%, 1/8 W

FIGURE 5. Component temperature test circuit.

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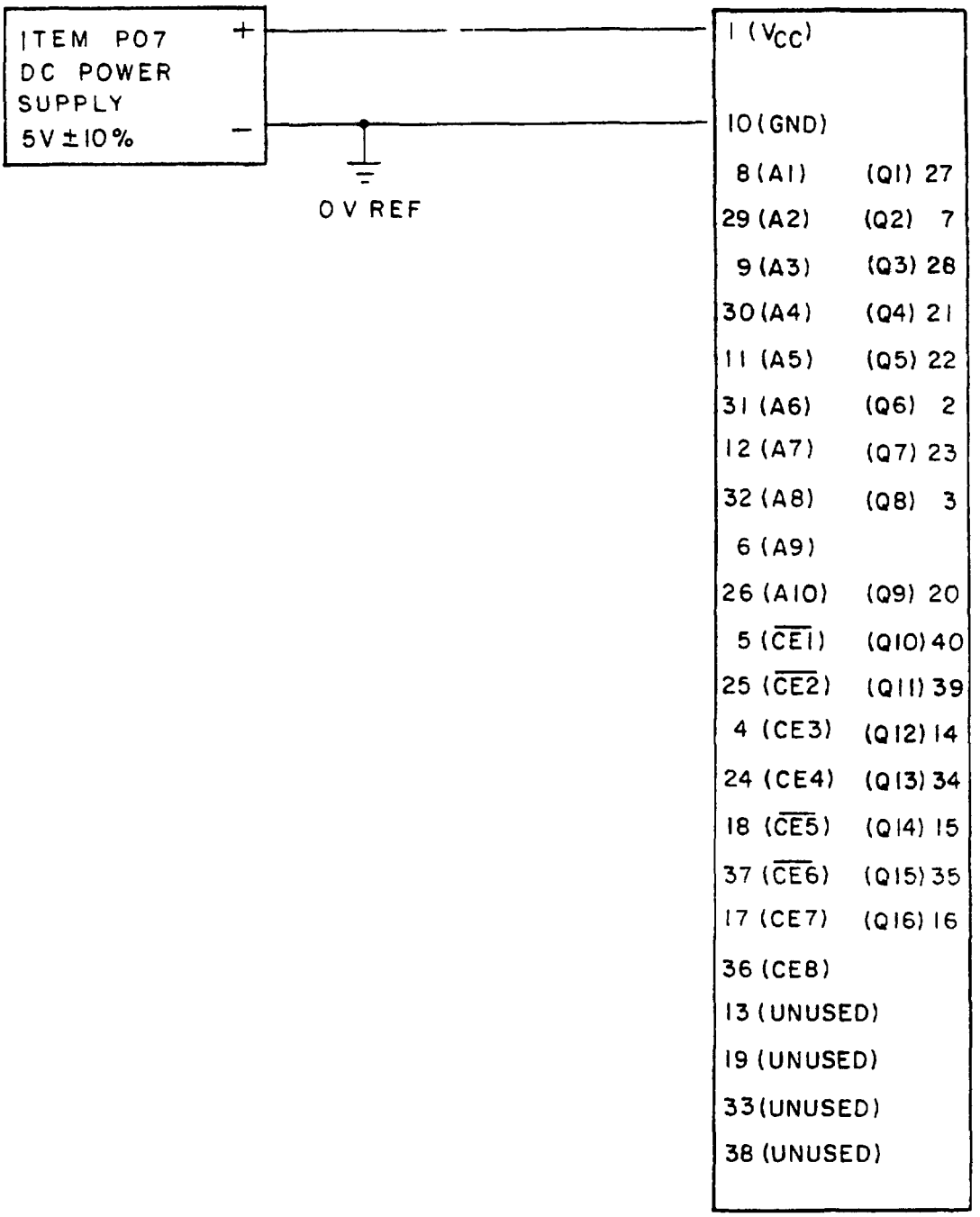


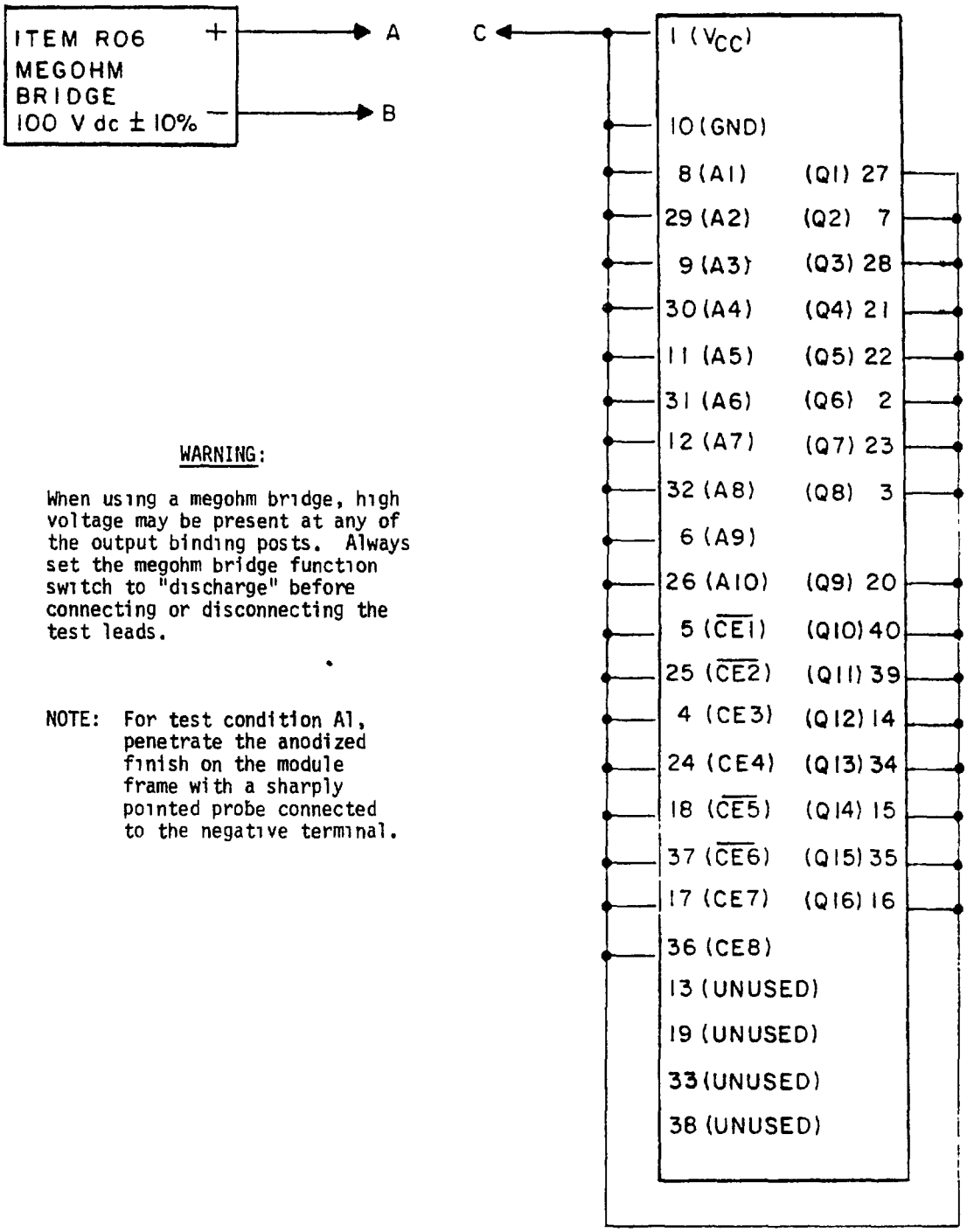
FIGURE 6. Vibration test circuit.

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* TABLE III. Isolation test.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
A1	Isolation	C,13,19,33,38	Frame	10	∞	MΩ	7
A2	Isolation, used to unused pins	C,13	19,33,38	"	"	"	"
A3	"	C,19	13,33,38	"	"	"	"
A4	"	C,33	13,19,38	"	"	"	"
A5	"	C,38	13,19,33	"	"	"	"

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WARNING:

When using a megohm bridge, high voltage may be present at any of the output binding posts. Always set the megohm bridge function switch to "discharge" before connecting or disconnecting the test leads.

NOTE: For test condition A1, penetrate the anodized finish on the module frame with a sharply pointed probe connected to the negative terminal.

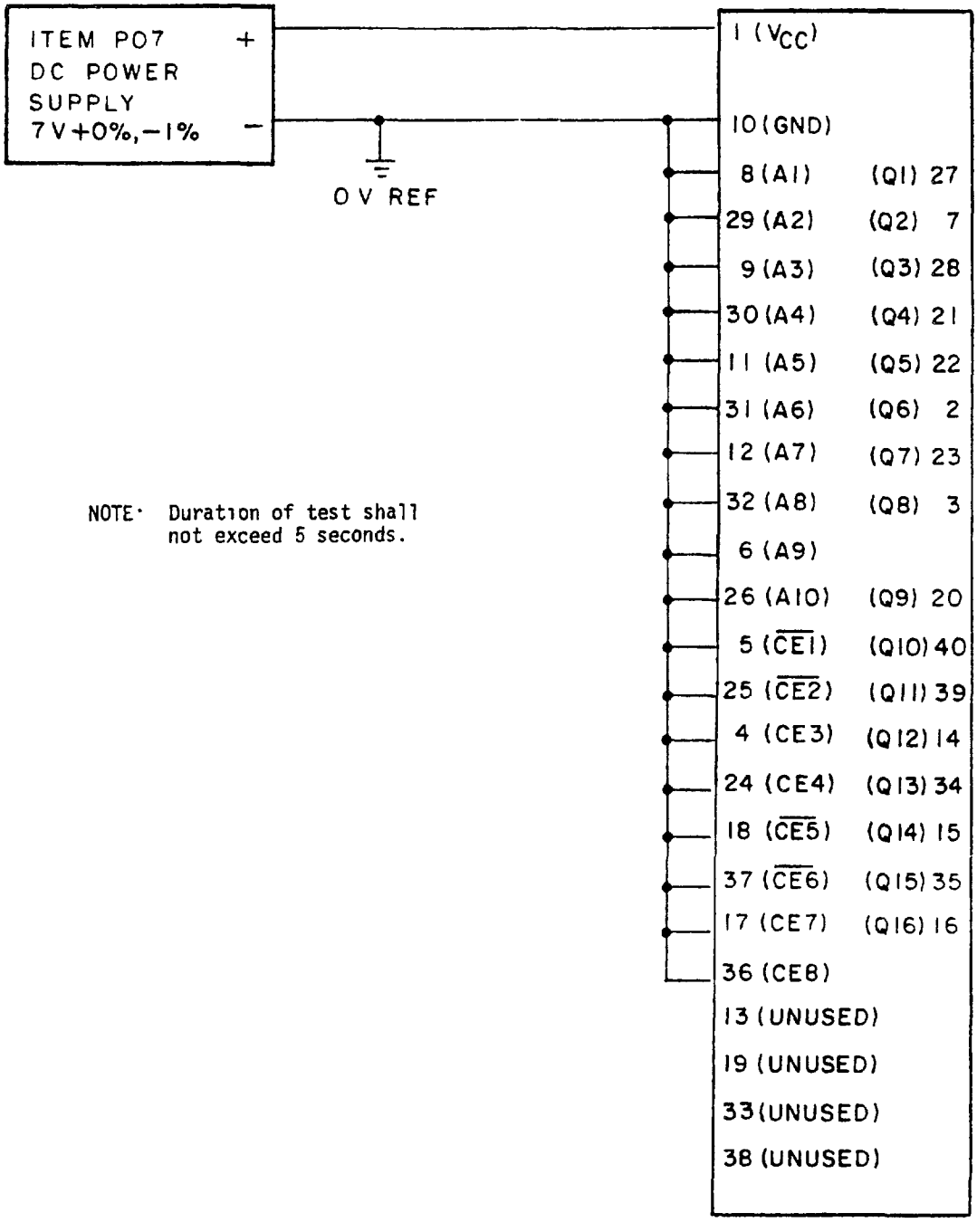
* FIGURE 7. Isolation test circuit for table III.

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TABLE IV. V_{CC} stress test.

Test condition	Parameter	Initial limits 25°C ±5°C ambient		Unit	Test figure
		Min	Max		
B	V_{CC} stress	Must pass remaining tests			8

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NOTE: Duration of test shall not exceed 5 seconds.

FIGURE 8. V_{CC} stress test circuit for table IV.

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TABLE V. Input breakdown current test.

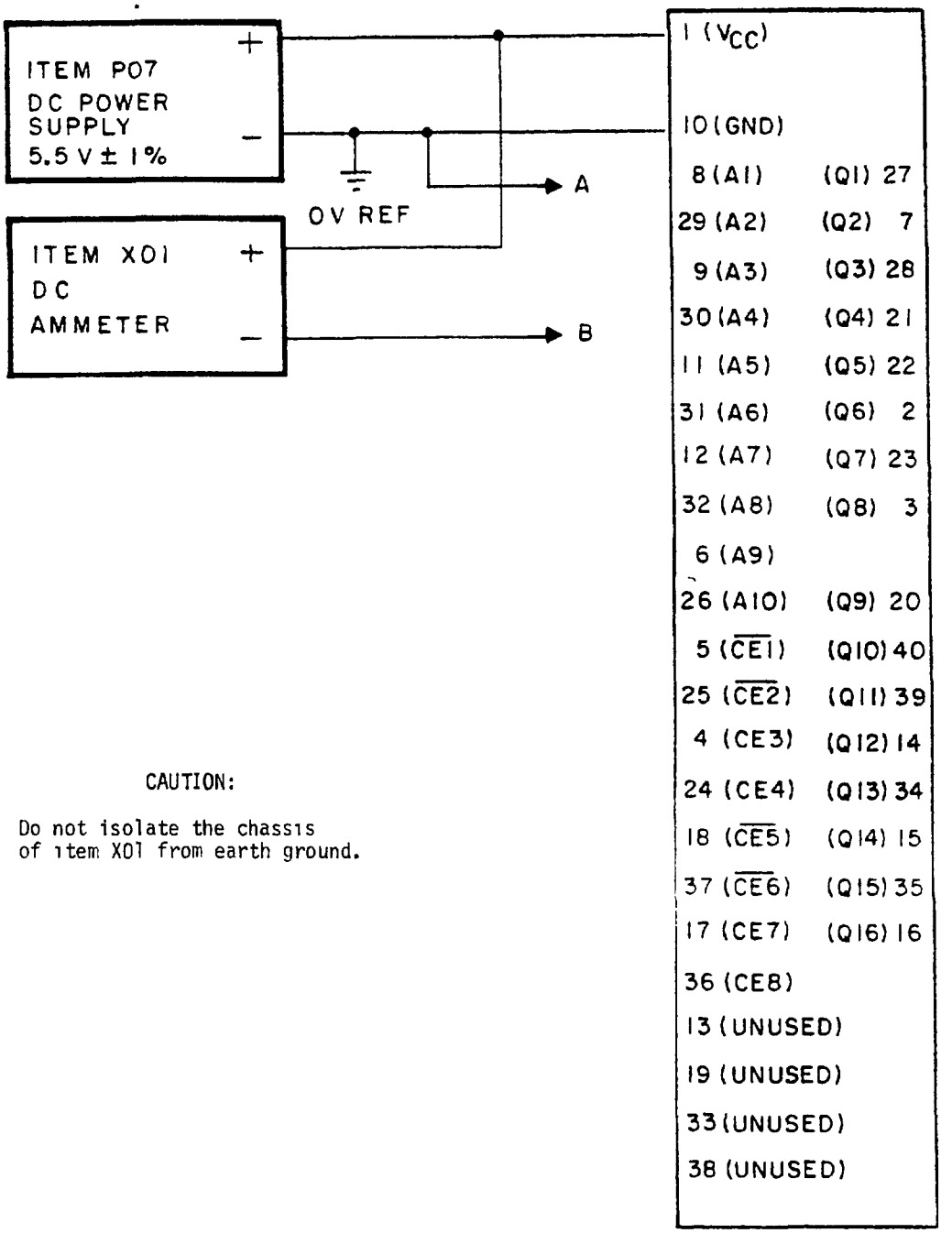
Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
C1	A1 Input breakdown current	4,5,6,9,11,12, 17,18,24,25,26, 29,30,31,32,36, 37	8	0	2	mA	9
C2	A2 "	4,5,6,8,9,11, 12,17,18,24, 25,26,30,32, 36,37	29	"	"	"	"
C3	A3 "	4,5,6,8,11,12, 17,18,24,25,26, 29,30,31,32,36, 37	9	"	"	"	"
C4	A4 "	4,5,6,8,9,11, 12,17,18,24,25, 26,29,31,32,36, 37	30	"	"	"	"
C5	A5 "	4,5,6,8,9,12, 17,18,24,25,26, 29,30,31,32,36, 37	11	"	"	"	"
C6	A6 "	4,5,6,8,9,11, 12,17,18,24,25, 26,29,30,32,36, 37	31	"	"	"	"
C7	A7 "	4,5,6,8,9,11, 17,18,24,25,26, 29,30,31,32,36, 37	12	"	"	"	"
C8	A8 "	4,5,6,8,9,11, 12,17,18,24,25, 26,29,30,31,36, 37	32	"	"	"	"
C9	A9 "	4,5,8,9,11,12, 17,18,24,25,26, 29,30,31,32,36, 37	6	"	"	"	"
C10	A10 "	4,5,6,8,9,11, 12,17,18,24,25, 29,30,31,32,36, 37	26	"	"	"	"

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TABLE V. Input breakdown current test - Continued.

Test condition	Parameter		Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	Min	Max		
C11	$\overline{CE1}$	Input breakdown current	4,6,8,9,11,12, 17,18,24,25,26, 29,30,31,32,36, 37	5	0	1	mA	9
C12	$\overline{CE2}$	"	4,5,6,8,9,11, 12,17,18,24,26, 29,30,31,32,36, 37	25	"	"	"	"
C13	CE3	"	5,6,8,9,11,12, 17,18,24,26,29, 30,31,32,36,37, 25	4	"	"	"	"
C14	CE4	"	4,5,6,8,9,11, 12,17,18,25,26, 29,30,31,32,36, 37	24	"	"	"	"
C15	$\overline{CE5}$	"	4,5,6,8,9,11, 12,17,24,25,26, 29,30,31,32,36, 37	18	"	"	"	"
C16	$\overline{CE6}$	"	4,5,6,8,9,11, 12,17,18,24,25, 26,29,30,31,32, 36	37	"	"	"	"
C17	CE7	"	4,5,6,8,9,11, 12,18,24,25,26, 29,30,31,32,36, 37	17	"	"	"	"
C18	CE8	"	4,5,6,8,9,11, 12,17,18,24,25, 26,29,30,31,32, 37	36	"	"	"	"

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CAUTION:

Do not isolate the chassis of item X01 from earth ground.

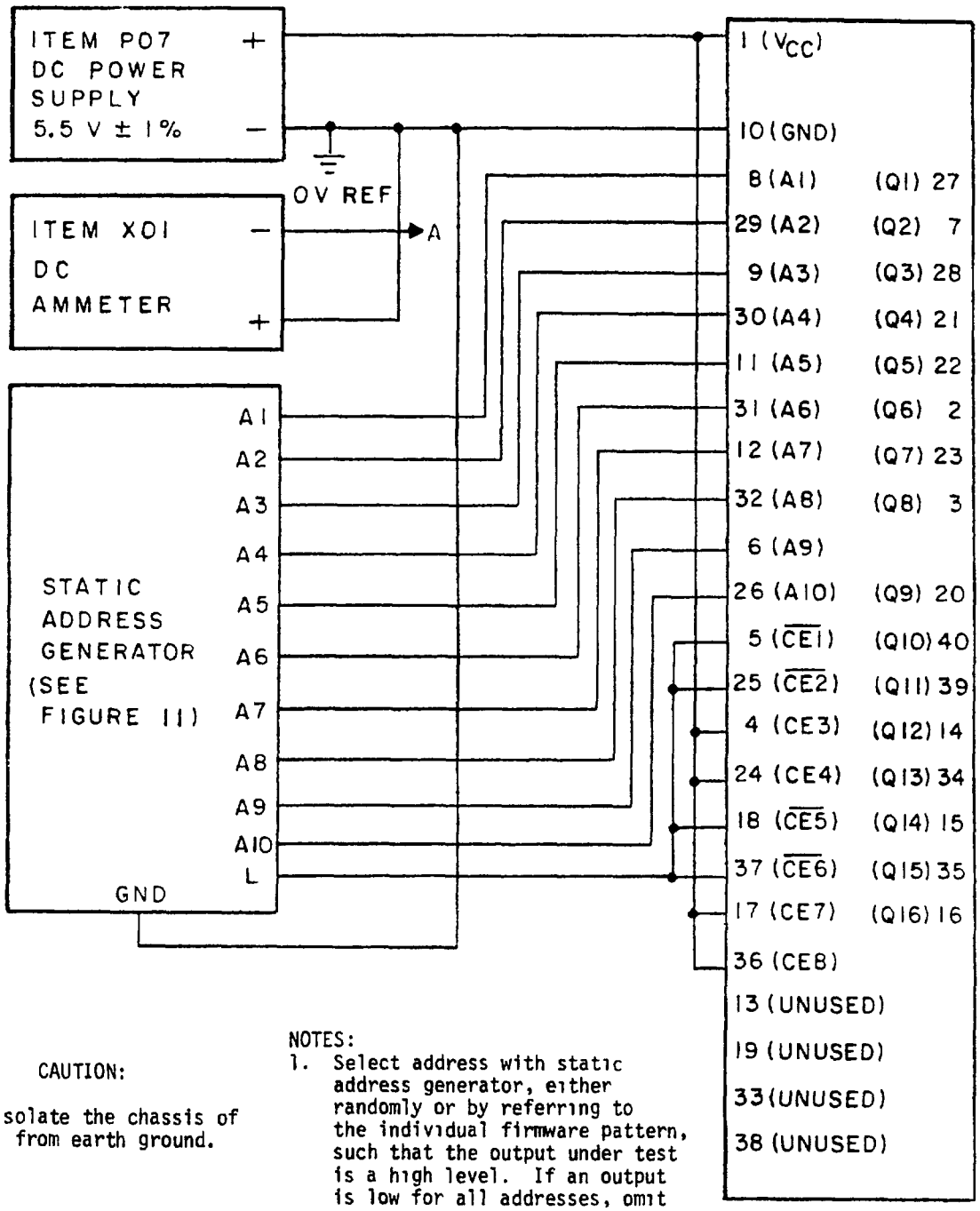
FIGURE 9. Input breakdown current test circuit for table V.

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TABLE VI. Output short circuit current test.

Test condition	Parameter	Connect	Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	Min	Max		
D1	Q1 Output short circuit current	27	-15	-100	mA	10
D2	Q2 "	7	"	"	"	"
D3	Q3 "	28	"	"	"	"
D4	Q4 "	21	"	"	"	"
D5	Q5 "	22	"	"	"	"
D6	Q6 "	2	"	"	"	"
D7	Q7 "	23	"	"	"	"
D8	Q8 "	3	"	"	"	"
D9	Q9 "	20	"	"	"	"
D10	Q10 "	40	"	"	"	"
D11	Q11 "	39	"	"	"	"
D12	Q12 "	14	"	"	"	"
D13	Q13 "	34	"	"	"	"
D14	Q14 "	15	"	"	"	"
D15	Q15 "	35	"	"	"	"
D16	Q16 "	16	"	"	"	"

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CAUTION:

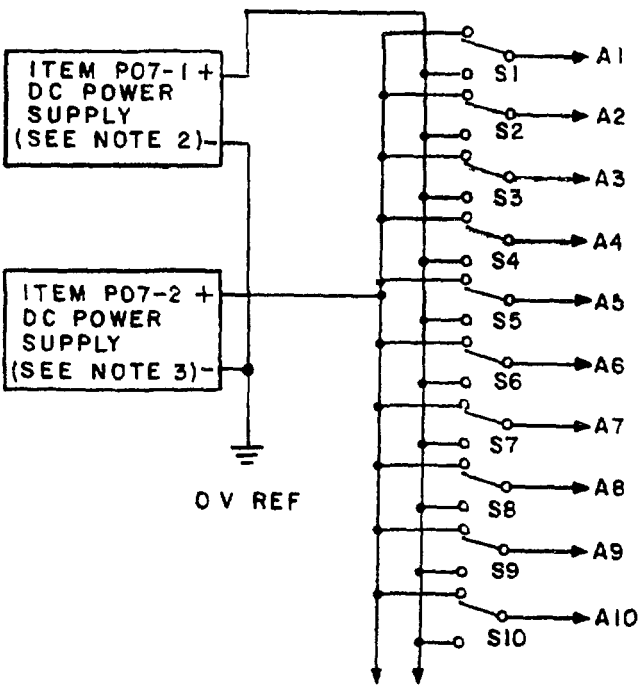
Do not isolate the chassis of item XO1 from earth ground.

NOTES:

1. Select address with static address generator, either randomly or by referring to the individual firmware pattern, such that the output under test is a high level. If an output is low for all addresses, omit this test for that output.
2. When performing the output short circuit current test, only one output shall be shorted at a time and the duration of the test shall not exceed ten seconds.

FIGURE 10. Output short circuit current test circuit for table VI.

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NOTES.
1. S1-S10 = SPDT switch.
2. For test conditions D, H, J, and N set H = 3.0 ±0.1 V. For test condition S set H = 2.0 +0.1, -0.0 V.
3. For test conditions D, H, J, and N set L = 0.0 V. For test condition S set L = 0.8 +0.0, -0.1 V.

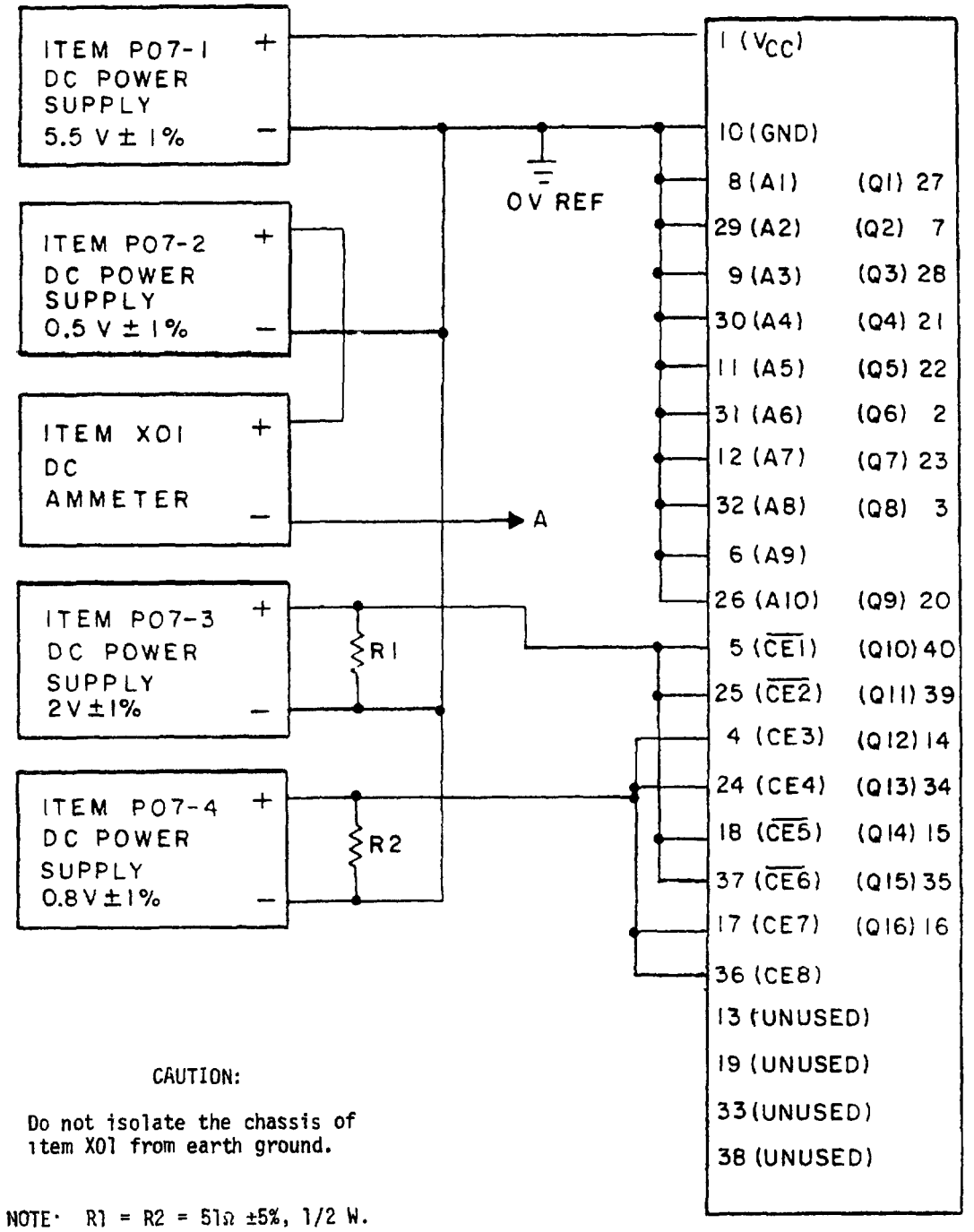
* FIGURE 11. Static address generator circuit.

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TABLE VII. Output leakage current (low-level) test.

Test condition	Parameter	Connect	Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	Min	Max		
E1	Q1 Output leakage current, low level	27	-100	100	µA	12
E2	Q2 "	7	"	"	"	"
E3	Q3 "	28	"	"	"	"
E4	Q4 "	21	"	"	"	"
E5	Q5 "	22	"	"	"	"
E6	Q6 "	2	"	"	"	"
E7	Q7 "	23	"	"	"	"
E8	Q8 "	3	"	"	"	"
E9	Q9 "	20	"	"	"	"
E10	Q10 "	40	"	"	"	"
E11	Q11 "	39	"	"	"	"
E12	Q12 "	14	"	"	"	"
E13	Q13 "	34	"	"	"	"
E14	Q14 "	15	"	"	"	"
E15	Q15 "	35	"	"	"	"
E16	Q16 "	16	"	"	"	"

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CAUTION:

Do not isolate the chassis of item XO1 from earth ground.

NOTE: R1 = R2 = 51Ω ±5%, 1/2 W.

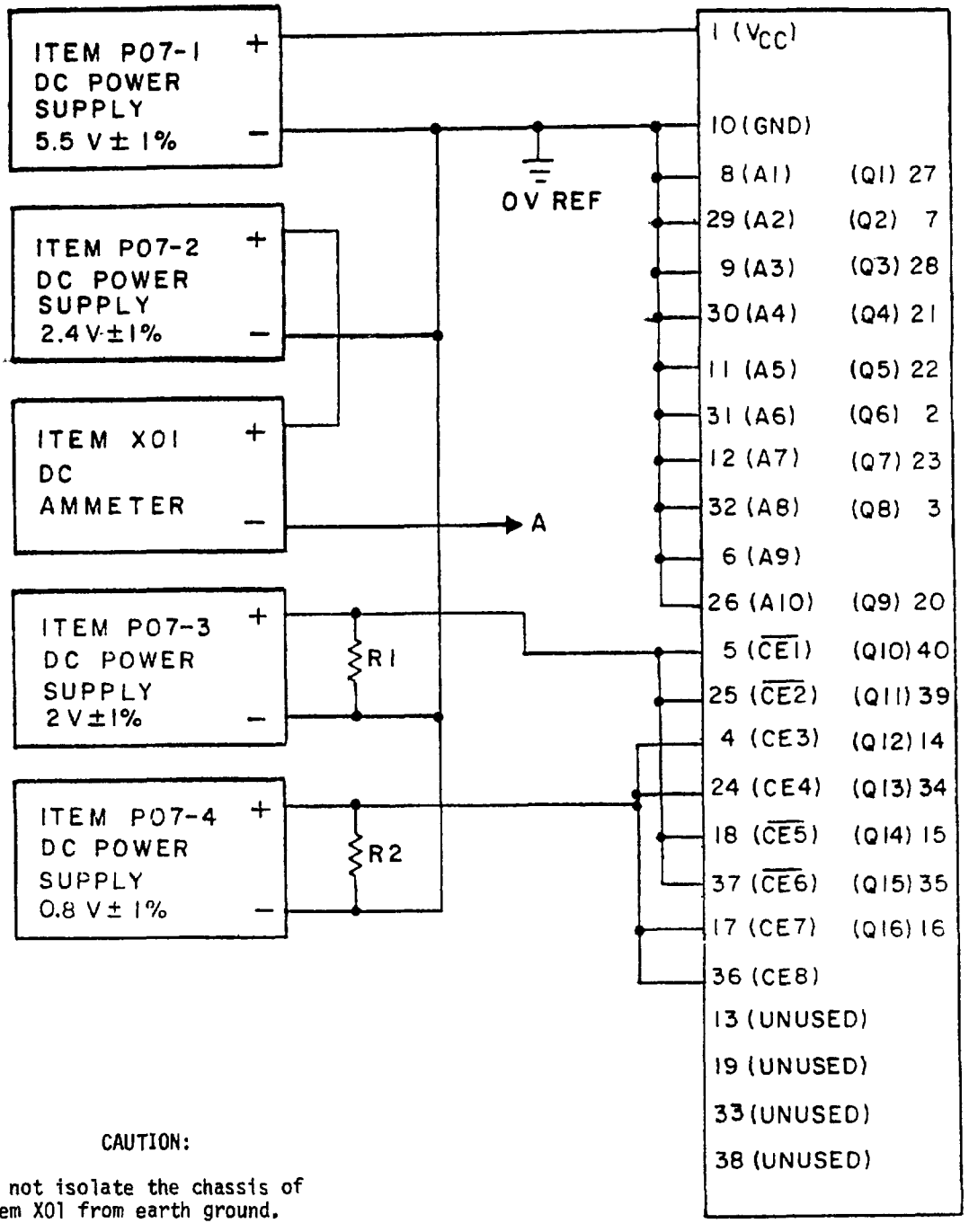
* FIGURE 12. Output leakage current (low-level) test circuit for table VII.

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TABLE VIII. Output leakage current (high-level) test.

Test condition	Parameter	Connect A to pin	Initial limits 25°C ±5°C ambient Min	Initial limits 25°C ±5°C ambient Max	Unit	Test figure
F1	Q1 Output leakage current, high level	27	-100	100	µA	13
F2	Q2 "	7	"	"	"	"
F3	Q3 "	28	"	"	"	"
F4	Q4 "	21	"	"	"	"
F5	Q5 "	22	"	"	"	"
F6	Q6 "	2	"	"	"	"
F7	Q7 "	23	"	"	"	"
F8	Q8 "	3	"	"	"	"
F9	Q9 "	20	"	"	"	"
F10	Q10 "	40	"	"	"	"
F11	Q11 "	39	"	"	"	"
F12	Q12 "	14	"	"	"	"
F13	Q13 "	34	"	"	"	"
F14	Q14 "	15	"	"	"	"
F15	Q15 "	35	"	"	"	"
F16	Q16 "	16	"	"	"	"

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CAUTION:

Do not isolate the chassis of item X01 from earth ground.

NOTE: R1 = R2 = 51Ω ±5%, 1/2 W.

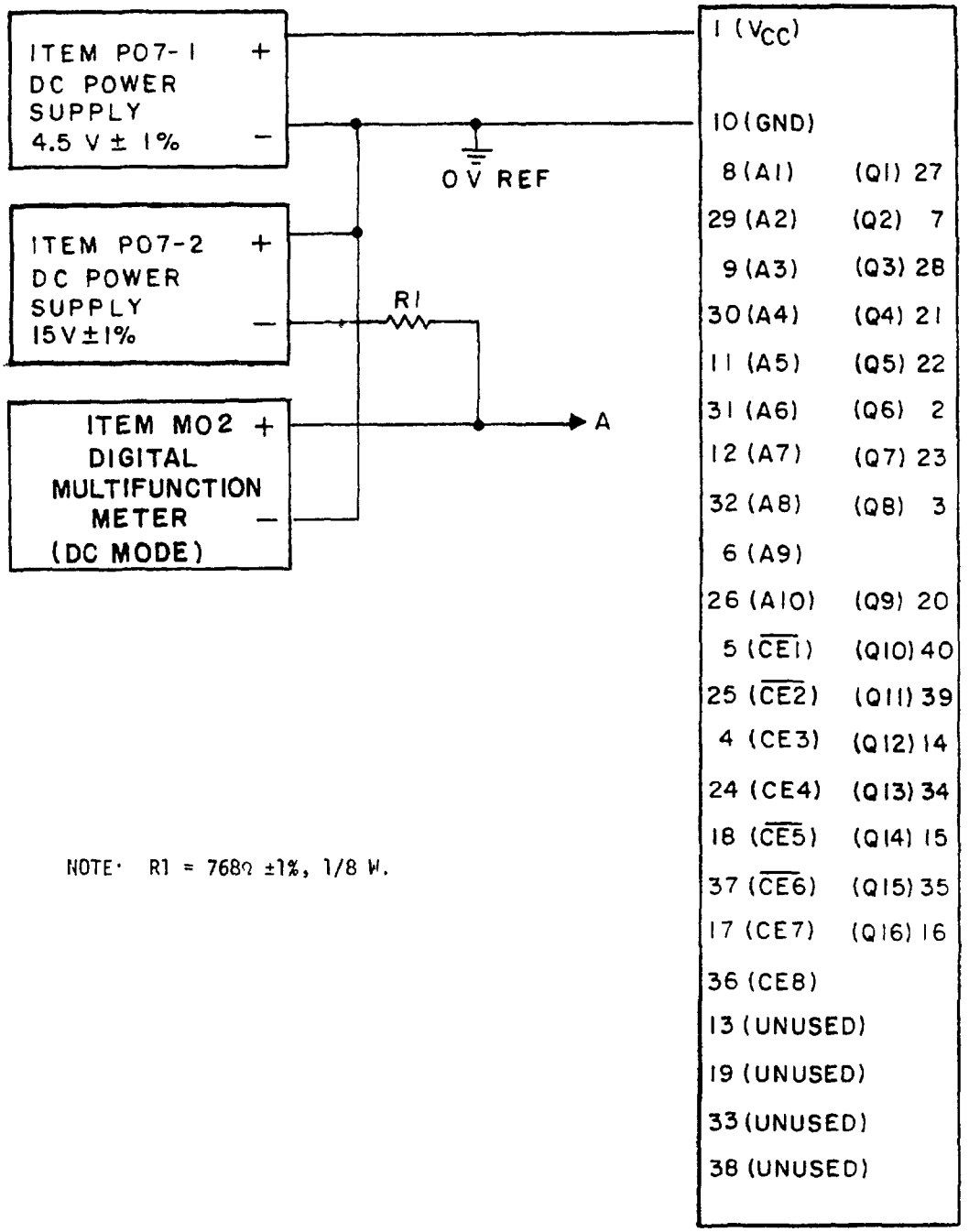
FIGURE 13. Output leakage current (high-level) test circuit for table VIII.

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TABLE IX. Input clamp diode voltage test.

Test condition	Parameter	Connect	Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	Min	Max		
G1	A1 Input clamp diode voltage	8	-0.4	-1.2	V	14
G2	A2 "	29	"	"	"	"
G3	A3 "	9	"	"	"	"
G4	A4 "	30	"	"	"	"
G5	A5 "	11	"	"	"	"
G6	A6 "	31	"	"	"	"
G7	A7 "	12	"	"	"	"
G8	A8 "	32	"	"	"	"
G9	A9 "	6	"	"	"	"
G10	A10 "	26	"	"	"	"
G11	$\overline{CE1}$ "	5	"	"	"	"
G12	$\overline{CE2}$ "	25	"	"	"	"
G13	CE3 "	4	"	"	"	"
G14	CE4 "	24	"	"	"	"
G15	$\overline{CE5}$ "	18	"	"	"	"
G16	$\overline{CE6}$ "	37	"	"	"	"
G17	CE7 "	17	"	"	"	"
G18	CE8 "	36	"	"	"	"

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NOTE: R1 = 768Ω ±1%, 1/8 W.

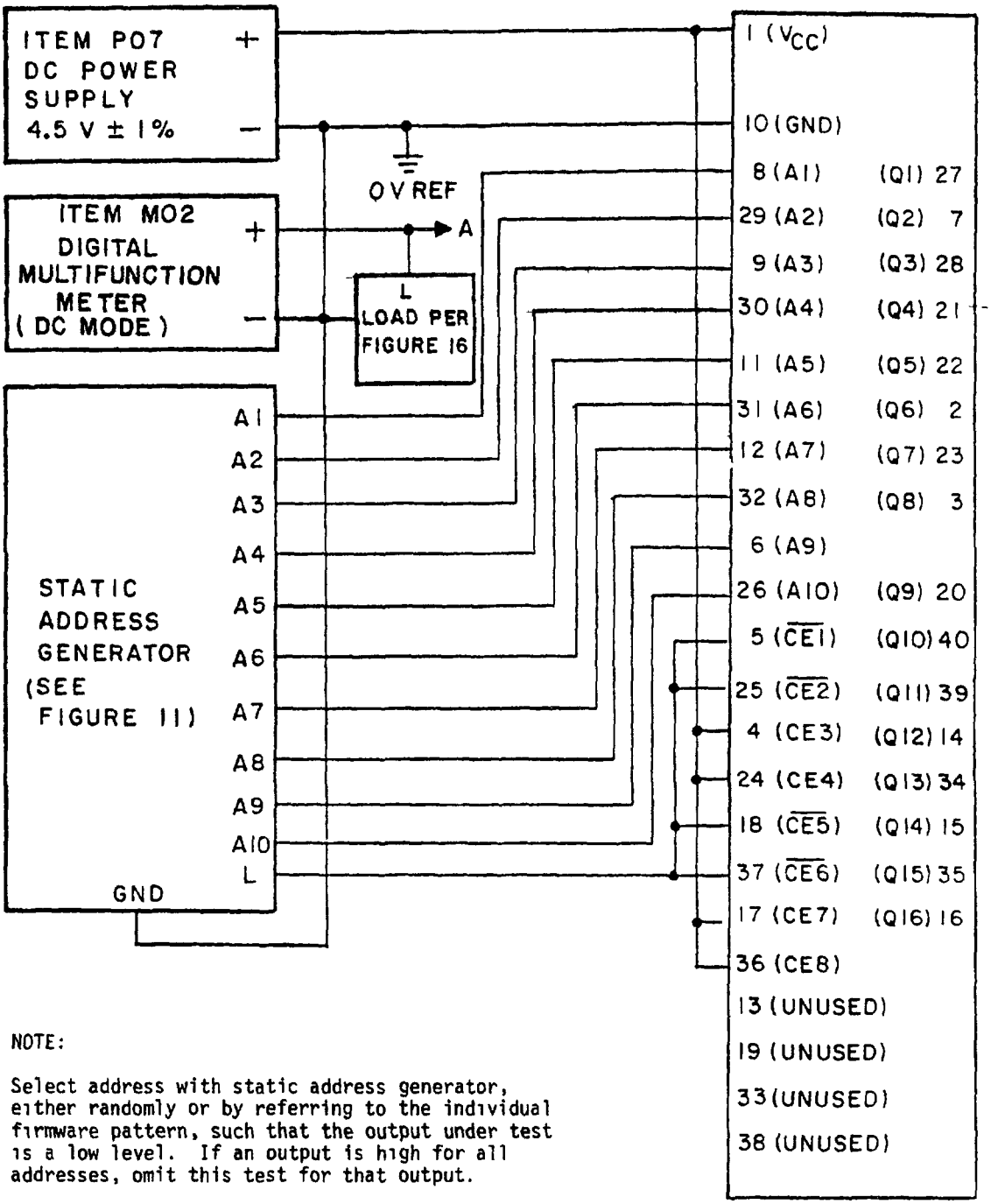
FIGURE 14. Input clamp diode voltage test circuit for table IX

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TABLE X. Output voltage (low-level) test.

Test condition	Parameter	Connect	Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	Min	Max		
H1	Q1 Output voltage, low level	27	0.0	0.5	V	15
H2	Q2 "	7	"	"	"	"
H3	Q3 "	28	"	"	"	"
H4	Q4 "	21	"	"	"	"
H5	Q5 "	22	"	"	"	"
H6	Q6 "	2	"	"	"	"
H7	Q7 "	23	"	"	"	"
H8 -	Q8 "	3	"	"	"	"
H9	Q9 "	20	"	"	"	"
H10	Q10 "	40	"	"	"	"
H11	Q11 "	39	"	"	"	"
H12	Q12 "	14	"	"	"	"
H13	Q13 "	34	"	"	"	"
H14	Q14 "	15	"	"	"	"
H15	Q15 "	35	"	"	"	"
H16	Q16 "	16	"	"	"	"

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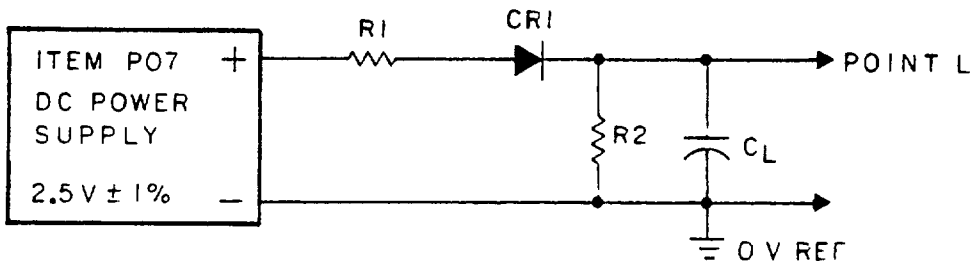


NOTE:

Select address with static address generator, either randomly or by referring to the individual firmware pattern, such that the output under test is a low level. If an output is high for all addresses, omit this test for that output.

FIGURE 15. Output voltage (low-level) test circuit for table X.

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NOTE:
1. CR1 = 1N3600, 1N4150, or equivalent.
2. R1 = Select value for a current flow of 16 mA ±5% with point L held at 0.5 V.
3. R2 = 1.2 kΩ ±1%, 1/8 W.
4. For test conditions N1-N64:
   $C_L \leq 10$ pF (including probe and fixture capacitance).
5. For test conditions P1, P2, R1-R16:
   $C_L = 50 \pm 5$ pF (including probe and fixture capacitance).
6. For test conditions P3, P4:
   $C_L = 300 \pm 15$ pF (including probe and fixture capacitance).
7. Capacitor C_L is not required for test conditions H, J, and S.
8. Lead length from load circuit to module output pin under test shall be less than 1 inch (25.4 mm).

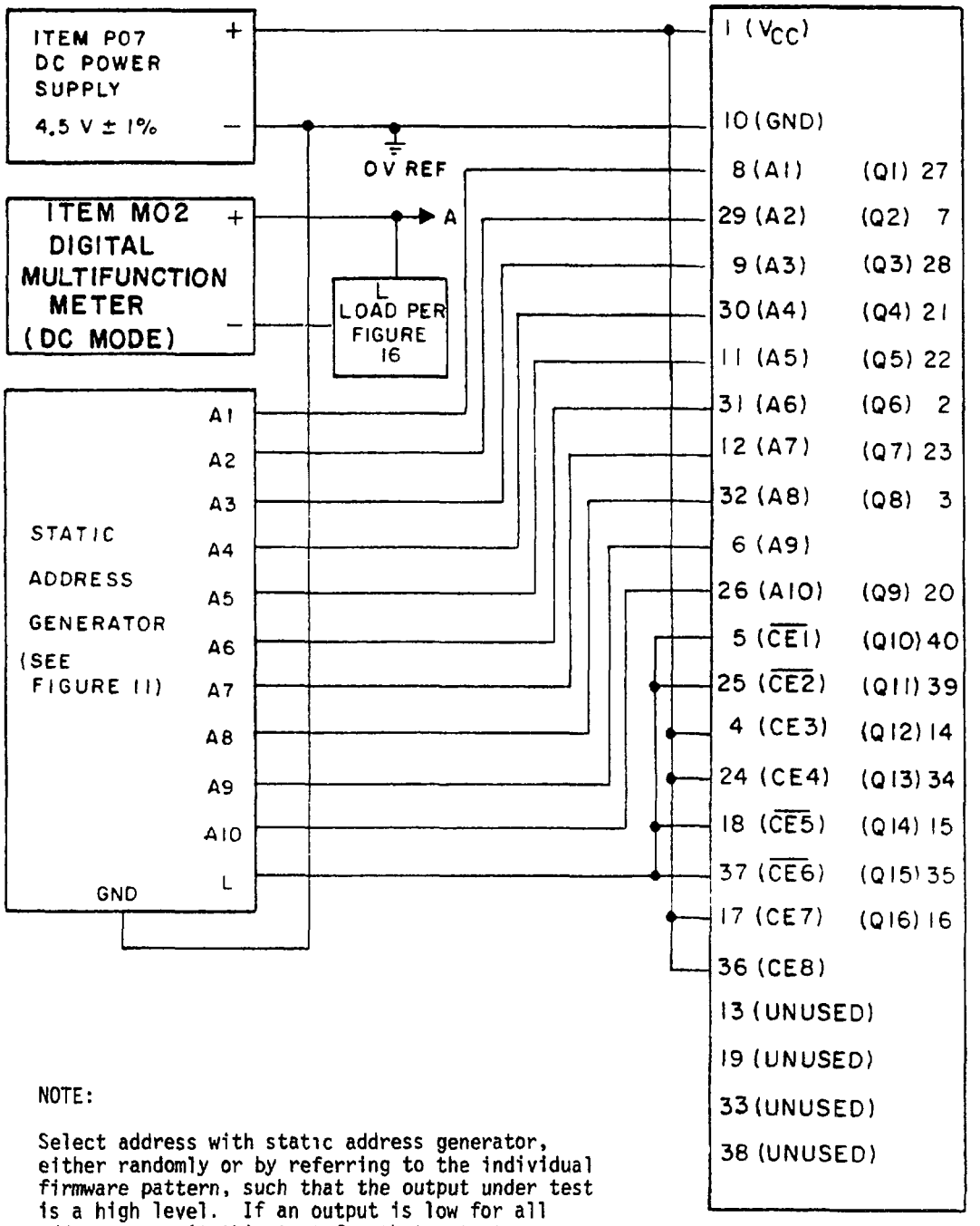
FIGURE 16. Logic load circuit.

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TABLE XI. Output voltage (high-level) test.

Test condition	Parameter	Connect A to pin	Initial limits 25°C ±5°C ambient Min	Initial limits 25°C ±5°C ambient Max	Unit	Test figure
J1	Q1 Output voltage, high level	27	2.4	4.5	V	17
J2	Q2 "	7	"	"	"	"
J3	Q3 "	28	"	"	"	"
J4	Q4 "	21	"	"	"	"
J5	Q5 "	22	"	"	"	"
J6	Q6 "	2	"	"	"	"
J7	Q7 "	23	"	"	"	"
J8	Q8 "	3	"	"	"	"
J9	Q9 "	20	"	"	"	"
J10	Q10 "	40	"	"	"	"
J11	Q11 "	39	"	"	"	"
J12	Q12 "	14	"	"	"	"
J13	Q13 "	34	"	"	"	"
J14	Q14 "	15	"	"	"	"
J15	Q15 "	35	"	"	"	"
J16	Q16 "	16	"	"	"	"

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NOTE:

Select address with static address generator, either randomly or by referring to the individual firmware pattern, such that the output under test is a high level. If an output is low for all addresses, omit this test for that output.

FIGURE 17. Output voltage (high-level) test circuit for table XI.

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TABLE XII. Input current (high-level) test.

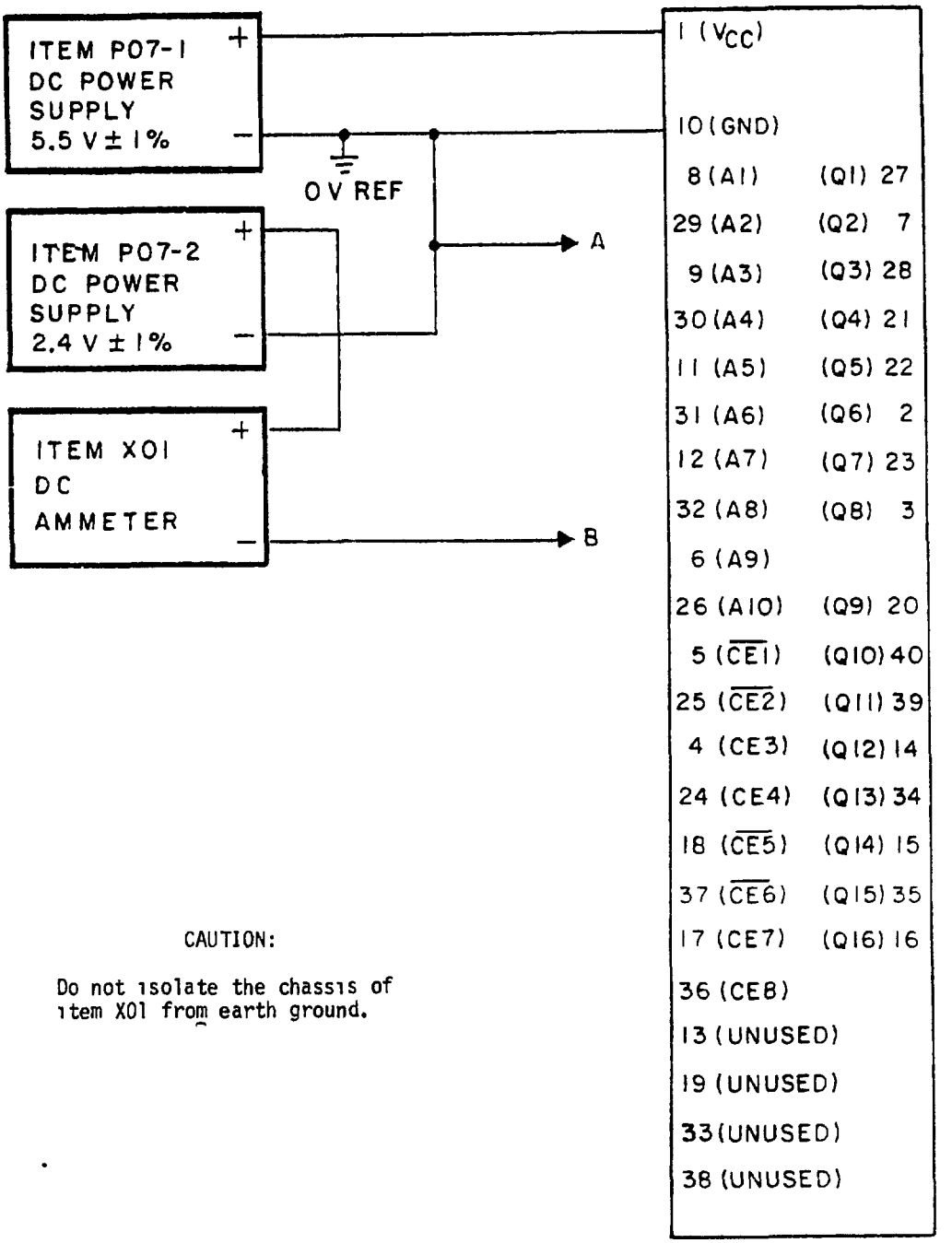
Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
K1	A1 Input current, high level	4,5,6,9,11,12, 17,18,24,25,26, 29,30,31,32,36, 37	8	0	100	µA	18
K2	A2 "	4,5,6,8,9,11, 12,17,18,24, 25,26,30,31,32, 36,37	29	"	"	"	"
K3	A3 "	4,5,6,8,11,12, 17,18,24,25,26, 29,30,31,32,36, 37	9	"	"	"	"
K4	A4 "	4,5,6,8,9,11, 12,17,18,24,25, 26,29,31,32,36, 37	30	"	"	"	"
K5	A5 "	4,5,6,8,9,12, 17,18,24,25,26, 29,30,31,32,36, 37	11	"	"	"	"
K6	A6 "	4,5,6,8,9,11, 12,17,18,24,25, 26,29,30,32,36, 37	31	"	"	"	"
K7	A7 "	4,5,6,8,9,11, 17,18,24,25,26, 29,30,31,32,36, 37	12	"	"	"	"
K8	A8 "	4,5,6,8,9,11, 12,17,18,24,25, 26,29,30,31,36, 37	32	"	"	"	"
K9	A9 "	4,5,8,9,11,12, 17,18,24,25,26, 29,30,31,32,36, 37	6	"	"	"	"
K10	A10 "	4,5,6,8,9,11, 12,17,18,24,25, 29,30,31,32,36, 37	26	"	"	"	"

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TABLE XII. Input current (high-level) test - Continued.

Test condition	Parameter		Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	Min	Max		
K11	$\overline{CE1}$	Input current, high level	4,6,8,9,11,12, 17,18,24,25,26, 29,30,31,32,36, 37	5	0	50	µA	18
K12	$\overline{CE2}$	"	4,5,6,8,9,11, 12,17,18,24,26, 29,30,31,32,36, 37	25	"	"	"	"
K13	CE3	"	5,6,8,9,11,12, 17,18,24,26,29, 30,31,32,36,37, 25	4	"	"	"	"
K14	CE4	"	4,5,6,8,9,11, 12,17,18,25,26, 29,30,31,32,36, 37	24	"	"	"	"
K15	$\overline{CE5}$	"	4,5,6,8,9,11, 12,17,24,25,26, 29,30,31,32,36, 37	18	"	"	"	"
K16	$\overline{CE6}$	"	4,5,6,8,9,11, 12,17,18,24,25, 26,29,30,31,32, 36	37	"	"	"	"
K17	CE7	"	4,5,6,8,9,11, 12,18,24,25,26, 29,30,31,32,36, 37	17	"	"	"	"
K18	CE8	"	4,5,6,8,9,11, 12,17,18,24,25, 26,29,30,31,32, 37	36	"	"	"	"

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CAUTION:

Do not isolate the chassis of item X01 from earth ground.

FIGURE 18. Input current (high-level) test circuit for table XII.

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TABLE XIII. Input current (low-level) test.

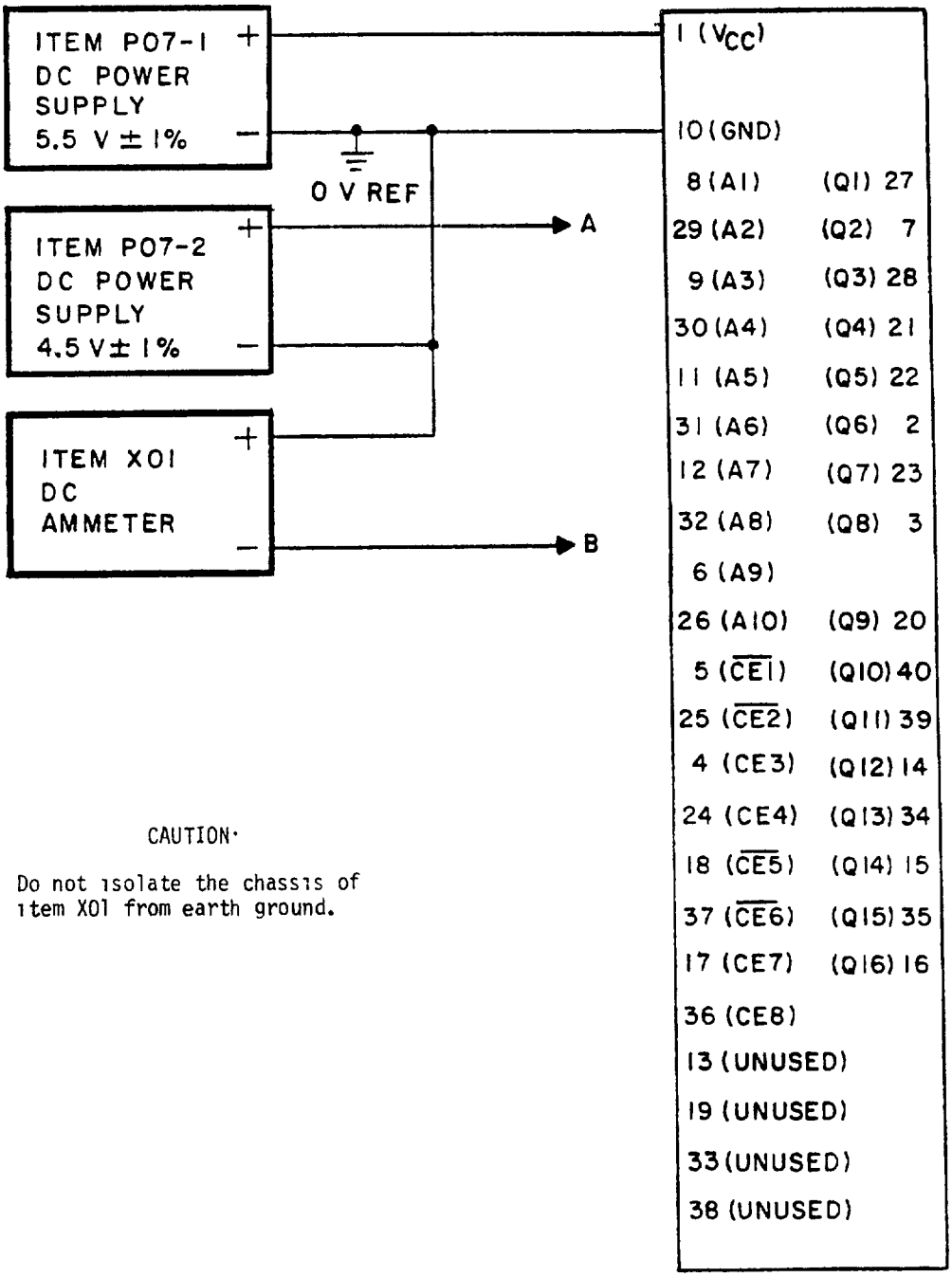
Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
L1	A1 Input current, low level	4,5,6,9,11,12, 17,18,24,25,26, 29,30,31,32,36, 37	8	0	-550	µA	19
L2	A2 "	4,5,6,8,9,11, 12,17,18,24, 25,26,30,31,32, 36,37	29	"	"	"	"
L3	A3 "	4,5,6,8,11,12, 17,18,24,25,26, 29,30,31,32,36, 37	9	"	"	"	"
L4	A4 "	4,5,6,8,9,11, 12,17,18,24,25, 26,29,31,32,36, 37	30	"	"	"	"
L5	A5 "	4,5,6,8,9,12, 17,18,24,25,26, 29,30,31,32,36, 37	11	"	"	"	"
L6	A6 "	4,5,6,8,9,11, 12,17,18,24,25, 26,29,30,32,36, 37	31	"	"	"	"
L7	A7 "	4,5,6,8,9,11, 17,18,24,25,26, 29,30,31,32,36, 37	12	"	"	"	"
L8	A8 "	4,5,6,8,9,11, 12,17,18,24,25, 26,29,30,31,36, 37	32	"	"	"	"
L9	A9 "	4,5,8,9,11,12, 17,18,24,25,26, 29,30,31,32,36, 37	6	"	"	"	"
L10	A10 "	4,5,6,8,9,11, 12,17,18,24,25, 29,30,31,32,36, 37	26	"	"	"	"

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TABLE XIII. Input current (low-level) test - Continued.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
L11	$\overline{CE1}$ Input current, low level	4,6,8,9,11,12, 17,18,24,25,26, 29,30,31,32,36, 37	5	0	-275	µA	19
L12	$\overline{CE2}$ "	4,5,6,8,9,11, 12,17,18,24,26, 29,30,31,32,36, 37	25	"	"	"	"
L13	CE3 "	5,6,8,9,11,12, 17,18,24,26,29, 30,31,32,36,37, 25	4	"	"	"	"
L14	CE4 "	4,5,6,8,9,11, 12,17,18,25,26, 29,30,31,32,36, 37	24	"	"	"	"
L15	$\overline{CE5}$ "	4,5,6,8,9,11, 12,17,24,25,26, 29,30,31,32,36, 37	18	"	"	"	"
L16	$\overline{CE6}$ "	4,5,6,8,9,11, 12,17,18,24,25, 26,29,30,31,32, 36	37	"	"	"	"
L17	CE7 "	4,5,6,8,9,11, 12,18,24,25,26, 29,30,31,32,36, 37	17	"	"	"	"
L18	CE8 "	4,5,6,8,9,11, 12,17,18,24,25, 26,29,30,31,32, 37	36	"	"	"	"

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CAUTION:

Do not isolate the chassis of item XO1 from earth ground.

FIGURE 19. Input current (low-level) test circuit for table XIII.

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TABLE XIV. Power supply current test.

Test condition	Parameter	Initial limits 25°C ±5°C ambient		Unit	Test figure
		Min	Max		
M	Power supply current	10	370	mA	20

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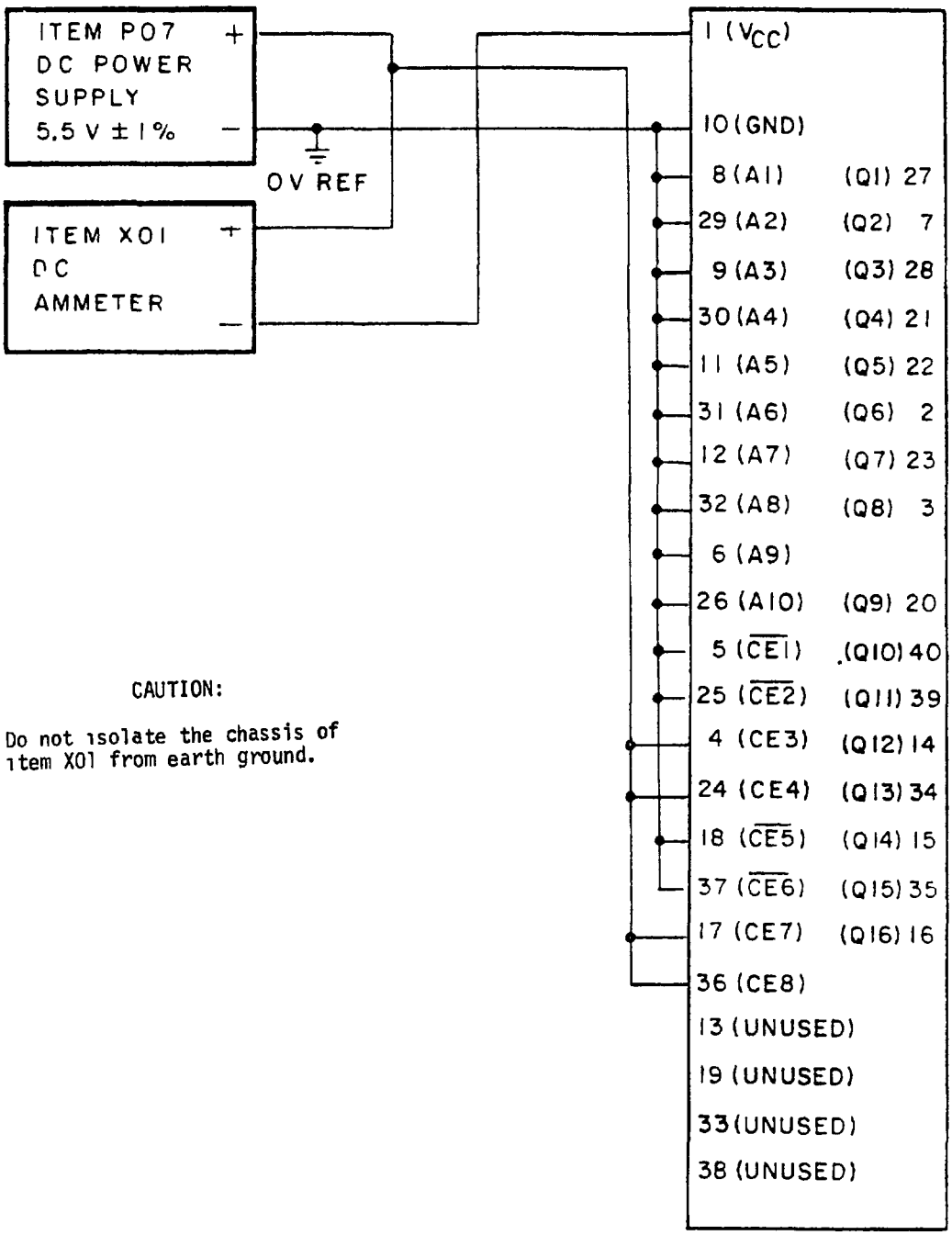


FIGURE 20. Power supply current test circuit for table XIV.

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TABLE XV. Enable recovery time (t_{ERL} and t_{ERH}) test.

Test condition	Parameter	Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
N1	Q1 Enable recovery time (t_{ERL}), C_L = ≤10 pF	4,24,18,37	5,25	17,36	27	1	50	ns	21,22
N2	Q2 "	"	"	"	7	"	"	"	"
N3	Q3 "	"	"	"	28	"	"	"	"
N4	Q4 "	"	"	"	21	"	"	"	"
N5	Q5 "	"	"	"	22	"	"	"	"
N6	Q6 "	"	"	"	2	"	"	"	"
N7	Q7 "	"	"	"	23	"	"	"	"
N8	Q8 "	"	"	"	3	"	"	"	"
N9	Q9 "	17,36,5,25	18,37	4,24	20	"	"	"	"
N10	Q10 "	"	"	"	40	"	"	"	"
N11	Q11 "	"	"	"	39	"	"	"	"
N12	Q12 "	"	"	"	14	"	"	"	"
N13	Q13 "	"	"	"	34	"	"	"	"
N14	Q14 "	"	"	"	15	"	"	"	"
N15	Q15 "	"	"	"	35	"	"	"	"
N16	Q16 "	"	"	"	16	"	"	"	"
N17	Q1 "	18,37	4,24	5,25,17,36	27	"	"	"	"
N18	Q2 "	"	"	"	7	"	"	"	"
N19	Q3 "	"	"	"	28	"	"	"	"
N20	Q4 "	"	"	"	21	"	"	"	"

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TABLE XV. Enable recovery time (t_{ERL} and t_{ERH}) test - Continued.

Test condition	Parameter	Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
N21	Q5 Enable recovery time (t_{ERL}), C_L = ≤10 pF	18,37	4,24	5,25,17,36	22	1	50	ns	21,22
N22	Q6 "	"	"	"	2	"	"	"	"
N23	Q7 "	"	"	"	23	"	"	"	"
N24	Q8 "	"	"	"	3	"	"	"	"
N25	Q9 "	5,25	17,36	18,37,4,24	20	"	"	"	"
N26	Q10 "	"	"	"	40	"	"	"	"
N27	Q11 "	"	"	"	39	"	"	"	"
N28	Q12 "	"	"	"	14	"	"	"	"
N29	Q13 "	"	"	"	34	"	"	"	"
N30	Q14 "	"	"	"	15	"	"	"	"
N31	Q15 "	"	"	"	35	"	"	"	"
N32	Q16 "	"	"	"	16	"	"	"	"
N33	Q1 Enable recovery time (t_{ERH}), C_L = ≤ 10 pF	4,24,18,37	5,25	17,36	27	"	"	"	"
N34	Q2 "	"	"	"	7	"	"	"	"
N35	Q3 "	"	"	"	28	"	"	"	"
N36	Q4 "	"	"	"	21	"	"	"	"
N37	Q5 "	"	"	"	22	"	"	"	"
N38	Q6 "	"	"	"	2	"	"	"	"
N39	Q7 "	"	"	"	23	"	"	"	"
N40	Q8 "	"	"	"	3	"	"	"	"

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TABLE XV. Enable recovery time (t_{ERL} and t_{ERH}) test - Continued.

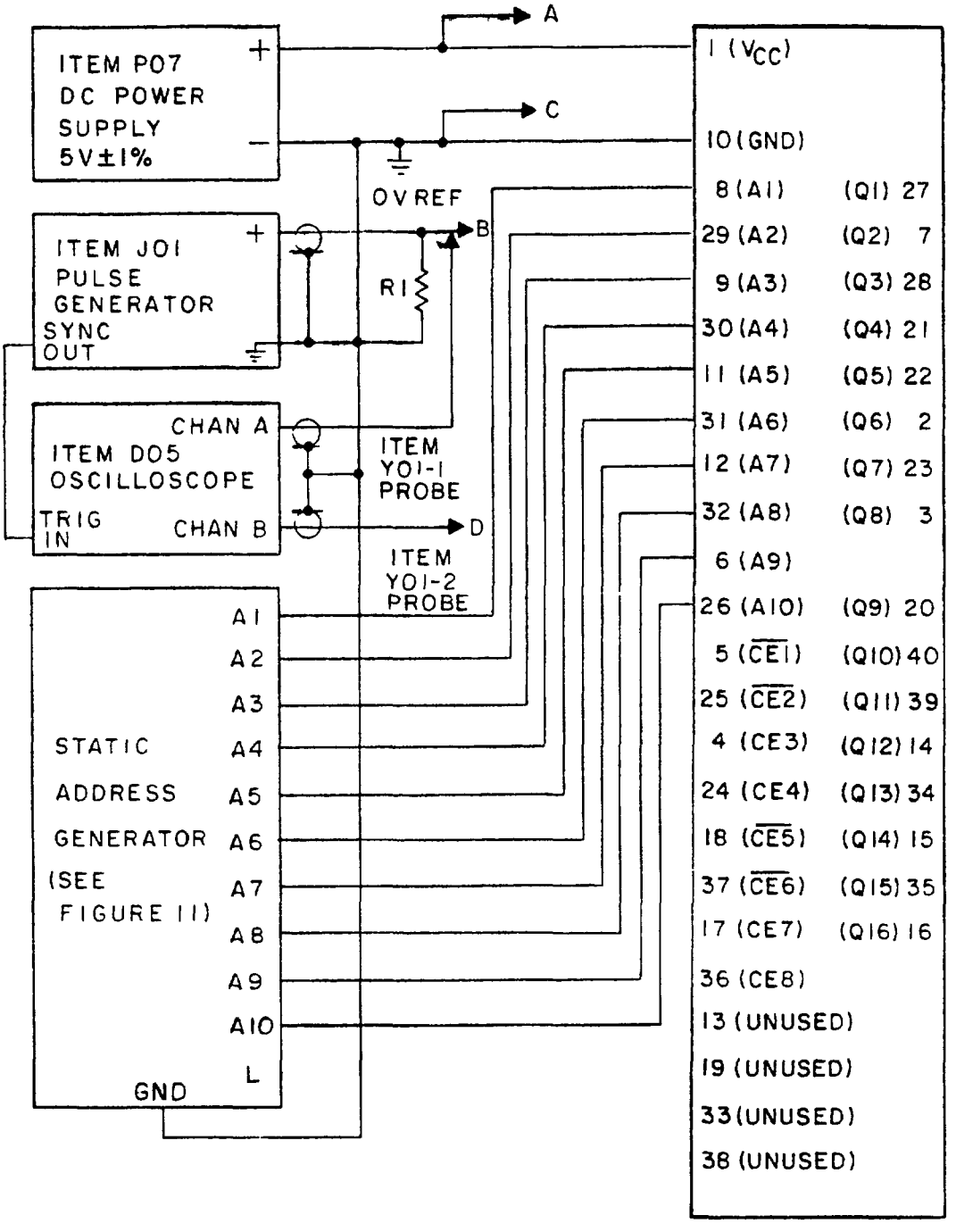
Test condition	Parameter	Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
N41	Q9 Enable recovery time (t_{ERH}), C_L = ≤10 pF	17,36,5,25	18,37	4,24	20	1	50	ns	21,22
N42	Q10 "	"	"	"	40	"	"	"	"
N43	Q11 "	"	"	"	39	"	"	"	"
N44	Q12 "	"	"	"	14	"	"	"	"
N45	Q13 "	"	"	"	34	"	"	"	"
N46	Q14 "	"	"	"	15	"	"	"	"
N47	Q15 "	"	"	"	35	"	"	"	"
N48	Q16 "	"	"	"	16	"	"	"	"
N49	Q1 "	18,37	4,24	5,25,17,36	27	"	"	"	"
N50	Q2 "	"	"	"	7	"	"	"	"
N51	Q3 "	"	"	"	28	"	"	"	"
N52	Q4 "	"	"	"	21	"	"	"	"
N53	Q5 "	"	"	"	22	"	"	"	"
N54	Q6 "	"	"	"	2	"	"	"	"
N55	Q7 "	"	"	"	23	"	"	"	"
N56	Q8 "	"	"	"	3	"	"	"	"
N57	Q9 "	5,25	17,36	18,37,4,24	20	"	"	"	"
N58	Q10 "	"	"	"	40	"	"	"	"
N59	Q11 "	"	"	"	39	"	"	"	"

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TABLE XV. Enable recovery time (t_{ERL} and t_{ERH}) test - Continued.

Test condition	Parameter	Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
N60	Q12 Enable recovery time (t_{ERH}), C_L = ≤10 pF	5,25	17,36	18,37,4,24	14	1	50	ns	21,22
N61	Q13 "	"	"	"	34	"	"	"	"
N62	Q14 "	"	"	"	15	"	"	"	"
N63	Q15 "	"	"	"	35	"	"	"	"
N64	Q16 "	"	"	"	16	"	"	"	"

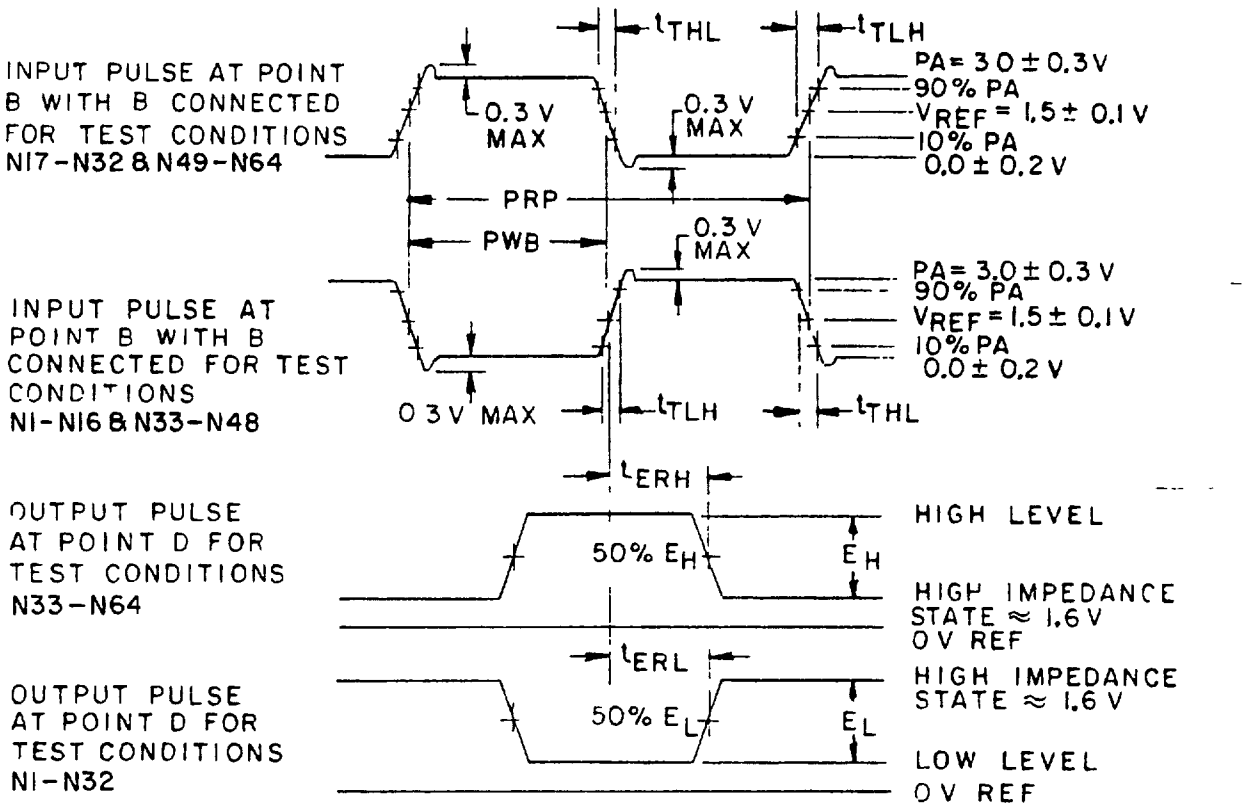
Source: https://assist.dla.mil -- Downloaded: 2026-02-20T17:41Z
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NOTES:
1. R1 = 51Ω ±5%, 1/4 W.
2. Load each output with the load specified on figure 16.

FIGURE 21. Enable recovery time (t_{ERL} and t_{ERH}) circuit for table XV.

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NOTES:
1. Input pulse parameters:
   PWB = 500 ±50 ns.
   PRP = 1,000 ±100 ns.
   $t_{TLH} = t_{THL}$ = 10 ±2 ns.
2. Select an address with the static address generator and one address input which will allow (when point B is connected) the output to change from a low level to the high impedance level (t_{ERL}) or a high level to the high impedance level (t_{ERH}). If an output has a high level for all addresses, omit the t_{ERL} test for that output, if an output has a low level for all addresses, omit the t_{ERH} test for that output.
3. When point B is connected to an enable pin, the static address generator output (L) is disconnected to the pin connected to point B.
4. Before making the measurements specified in table XV verify that the output pulse high level is $\geq$ 2.4 V and the low level is $\leq$ 0.5 V.
5. The lead length of electrical conductors between resistor R1 and input pin and between probes YO1-1 and YO1-2 and input pins shall be less than 1 inch (25.4 mm).
6. All voltages are referenced to ground.

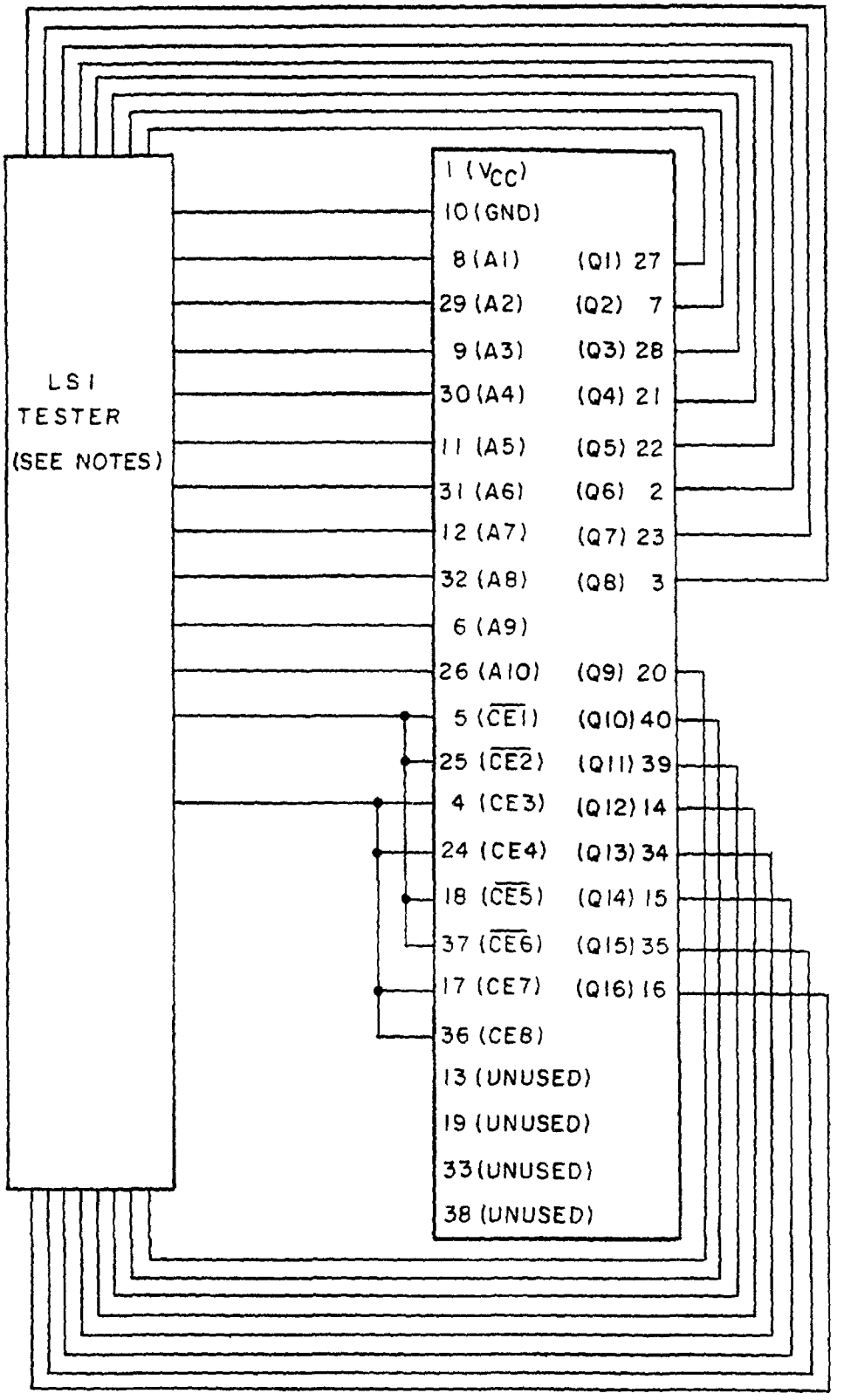
FIGURE 22. Enable recovery time (t_{ERL} and t_{ERH}) test circuit waveforms for table XV.

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TABLE XVI. Dynamic test.

Test condition	Parameter	Initial limits 25°C ±5°C ambient		Unit	Test figure
		Min	Max		
P1	Galloping read access test pattern, C_L = 50 pF, V_{CC} = 4.5 V	Module shall conform to figure 24 requirements (see notes 7 and 8)			23,24
P2	Galloping read access test pattern, C_L = 50 pF, V_{CC} = 5.5 V	"			"
P3	Galloping read access test pattern, C_L = 300 pF, V_{CC} = 4.5 V	"			"
P4	Galloping read access test pattern, C_L = 300 pF, V_{CC} = 5.5 V	"			"

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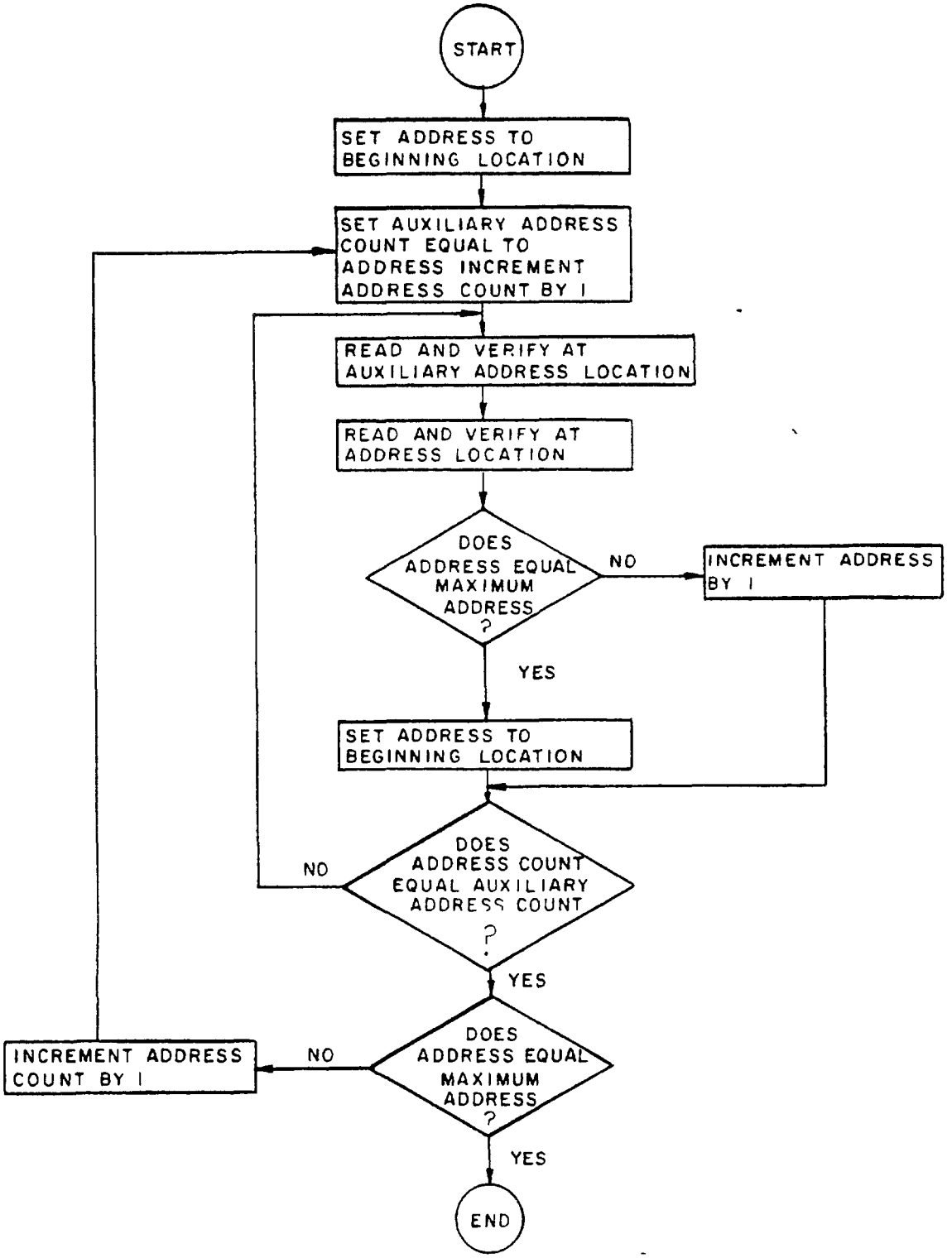
* FIGURE 23. Dynamic test circuit for table XVI.

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NOTES:
1. V_{CC} = 4.5 V ±1% for test conditions P1 and P3.
2. V_{CC} = 5.5 V ±1% for test conditions P2 and P4.
3. Load each output under test with the load specified on figure 16.

* FIGURE 23. Dynamic test circuit for table XVI - Continued.

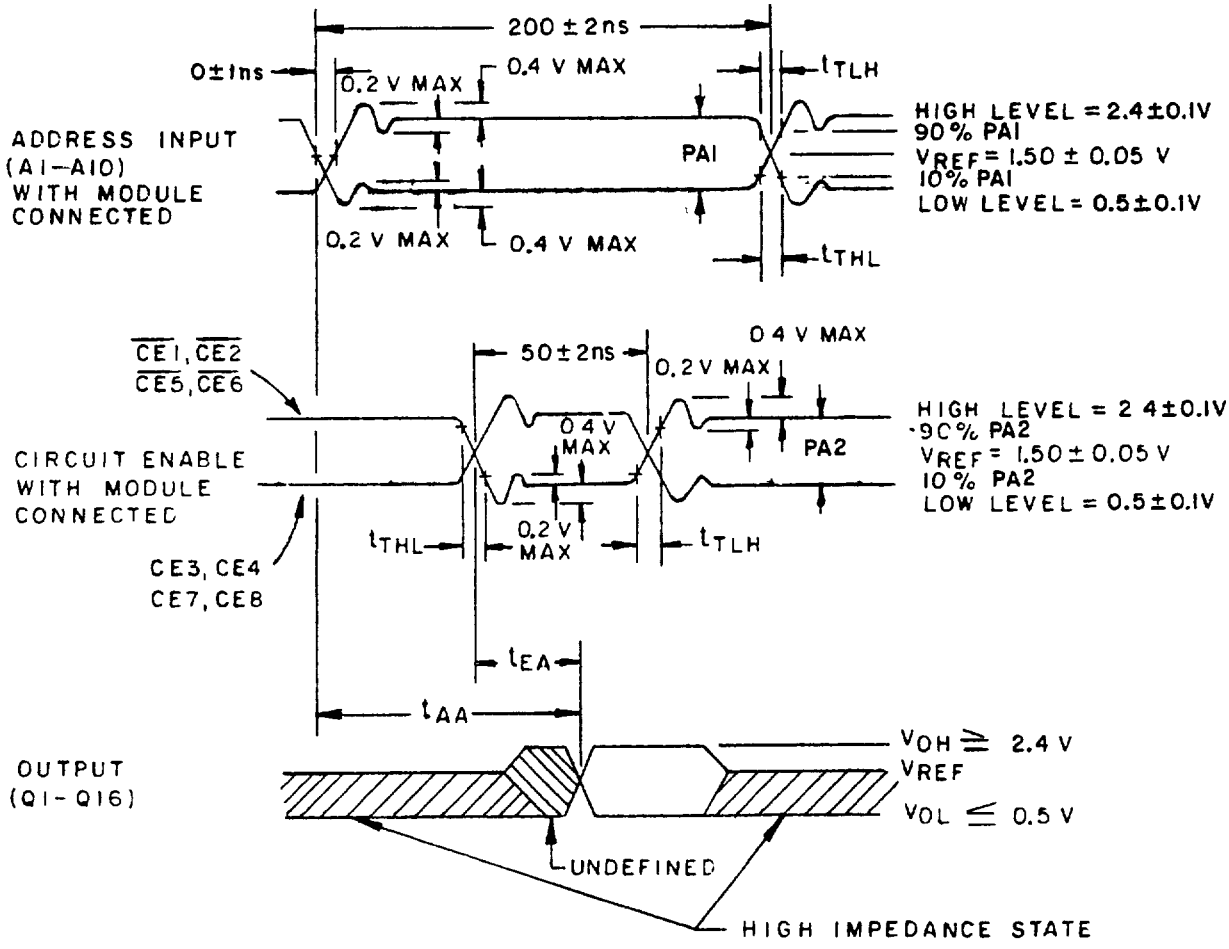
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NOTE: Output pattern is read and verified at each location.

FIGURE 24. Galloping read access test pattern program flow chart, waveforms, and program for table XVI.

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* FIGURE 24. Galloping read access test pattern program flow chart, waveforms, and program for table XVI - Continued.

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NOTES:

		Class 1	Class 2
1.	High impedance state = 1.6 V.		
2.	Address access time, C_L = 50 pF (t_{AA1}) =	70 $^{+0}_{-3}$ ns	80 $^{+0}_{-3}$ ns
3.	Address access time, C_L = 300 pF (t_{AA2}) =	100 $^{+0}_{-5}$ ns	110 $^{+0}_{-5}$ ns
4.	Enable access time, C_L = 50 pF (t_{EA1}) =	55 $^{+0}_{-2}$ ns	65 $^{+0}_{-2}$ ns
5.	Enable access time, C_L = 300 pF (t_{EA2}) =	75 $^{+0}_{-2}$ ns	85 $^{+0}_{-2}$ ns

6. $t_{TLH} = t_{THL} = 10 \pm 4$ ns.

7. The objective of the galloping read access test pattern is to test address access time. It is necessary to test all possible address transitions. The test also verifies circuit enable time (t_{EA}) and address access time (t_{AA}).

8. A program that fulfills the galloping read access test objective would be:

a. Read memory address zero.
b. Read memory address one.
c. Read memory address zero.
d. Read memory address two.
e. Read memory address zero.
f. Read memory address three.

•
•
•

g. Read memory address zero.
h. Read memory address 1024.
i. Read memory address one.
j. Read memory address two.
k. Read memory address one.
l. Read memory address three.

•
•
•

m. Read memory address one.
n. Read memory address 1024.
o. Read memory address one.
p. Read memory address zero.
q. Read memory address two.
r. Read memory address three.
s. Read memory address two.
t. Read memory address four.

•
•
•

u. Continue this pattern until all memory addresses have been tested to all other memory addresses.

* FIGURE 24. Galloping read access test pattern program flow chart, waveforms, and program for table XVI - Continued.

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TABLE XVII. Input crosstalk (low-level and high-level) test.

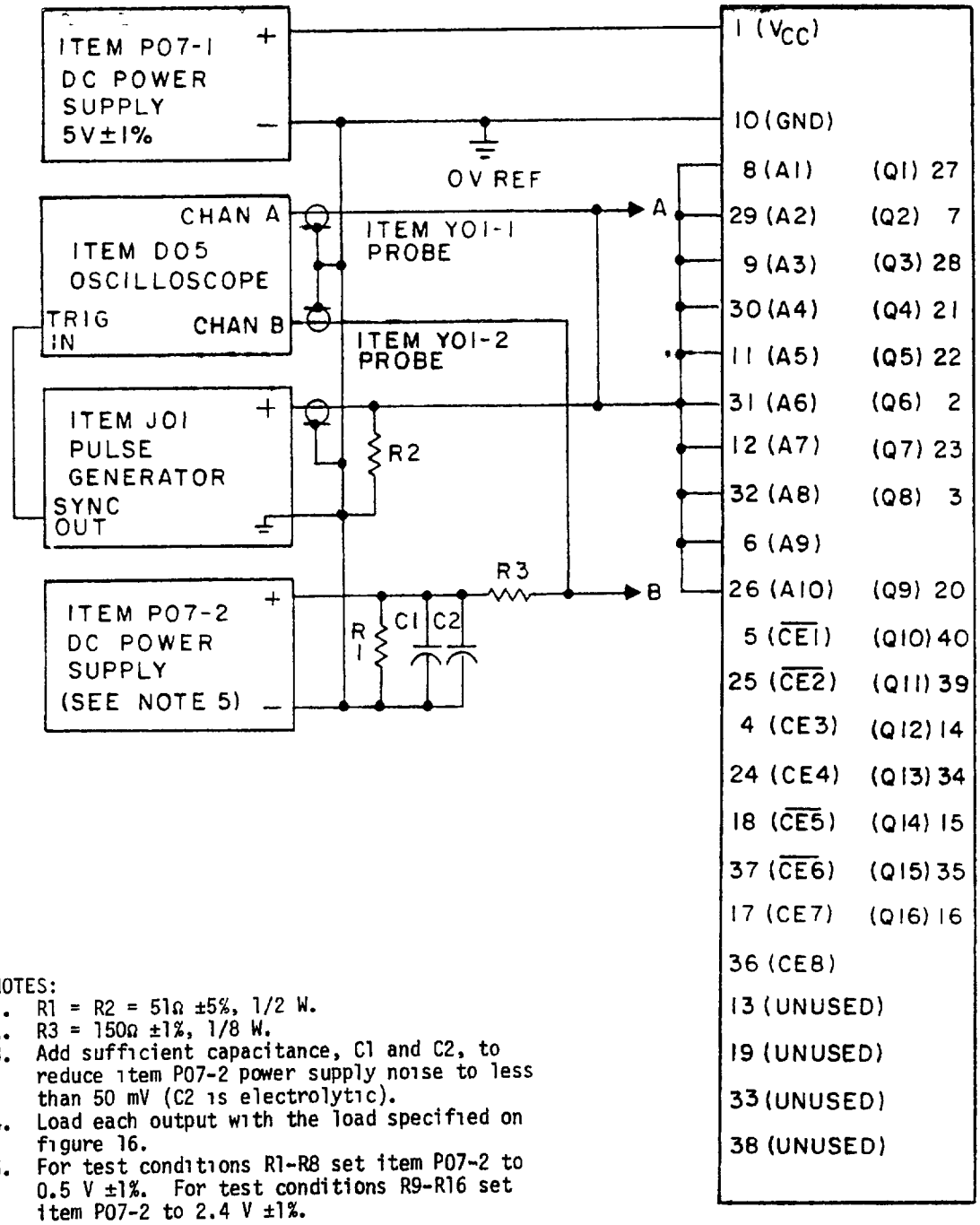
Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
R1	Input crosstalk, low level	25,4,24,18,37, 17,36	5	0	400	mV	25,26
R2	"	5,4,24,18,37, 17,36	25	"	"	"	"
R3	"	5,25,24,18,37, 17,36	4	"	"	"	"
R4	"	5,25,4,18,37, 17,36	24	"	"	"	"
R5	"	5,25,4,24,37, 17,36	18	"	"	"	"
R6	"	5,25,4,24,18, 17,36	37	"	"	"	"
R7	"	5,25,4,24,18, 37,36	17	"	"	"	"
R8	"	5,25,4,24,18, 37,17	36	"	"	"	"

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TABLE XVII. Input crosstalk (low-level and high-level) test - Continued.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
R9	Input crosstalk, high level	25,4,24,18,37, 17,36	5	0	400	mV	25,26
R10	"	5,4,24,18,37, 17,36	25	"	"	"	"
R11	"	5,25,24,18,37, 17,36	4	"	"	"	"
R12	"	5,25,4,18,37, 17,36	24	"	"	"	"
R13	"	5,25,4,24,37, 17,36	18	"	"	"	"
R14	"	5,25,4,24,18, 17,36	37	"	"	"	"
R15	"	5,25,4,24,18, 37,36	17	"	"	"	"
R16	"	5,25,4,24,18, 37,17	36	"	"	"	"

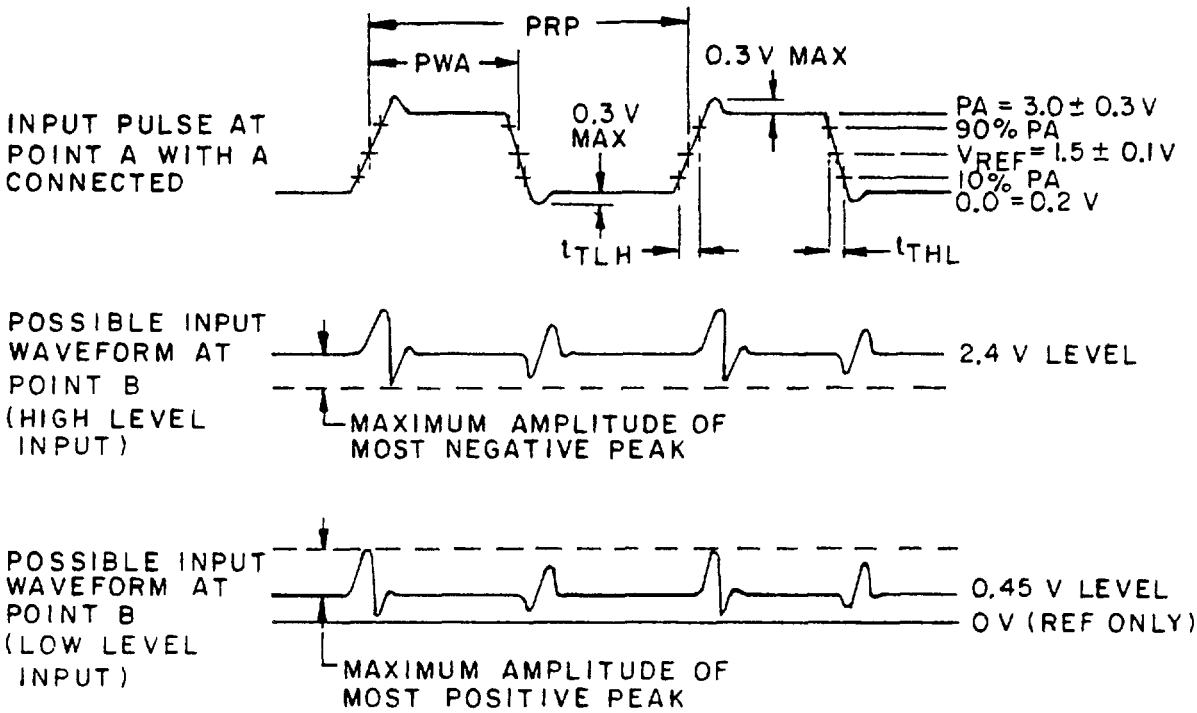
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NOTES:
1. R1 = R2 = 51Ω ±5%, 1/2 W.
2. R3 = 150Ω ±1%, 1/8 W.
3. Add sufficient capacitance, C1 and C2, to reduce item P07-2 power supply noise to less than 50 mV (C2 is electrolytic).
4. Load each output with the load specified on figure 16.
5. For test conditions R1-R8 set item P07-2 to 0.5 V ±1%. For test conditions R9-R16 set item P07-2 to 2.4 V ±1%.

FIGURE 25. Input crosstalk (low-level and high-level) test circuit for table XVII.

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NOTES:
1. Input pulse parameters:
   $t_{THL} = t_{TLH} = 20 \pm 2$ ns.
   PWA = 500 ±50 ns.
   PRP = 1,000 ±100 ns.
2. The lead length of electrical conductors between resistors R1 and R2 and input pins, probes Y01-1 and Y01-2 to input pins, load circuits to output pins, and ground leads of probes Y01-1 and Y01-2 to test fixture and module ground pins shall be less than 2 inches (50.8 mm).

FIGURE 26. Input crosstalk (low-level and high-level) test circuit waveforms for table XVII.

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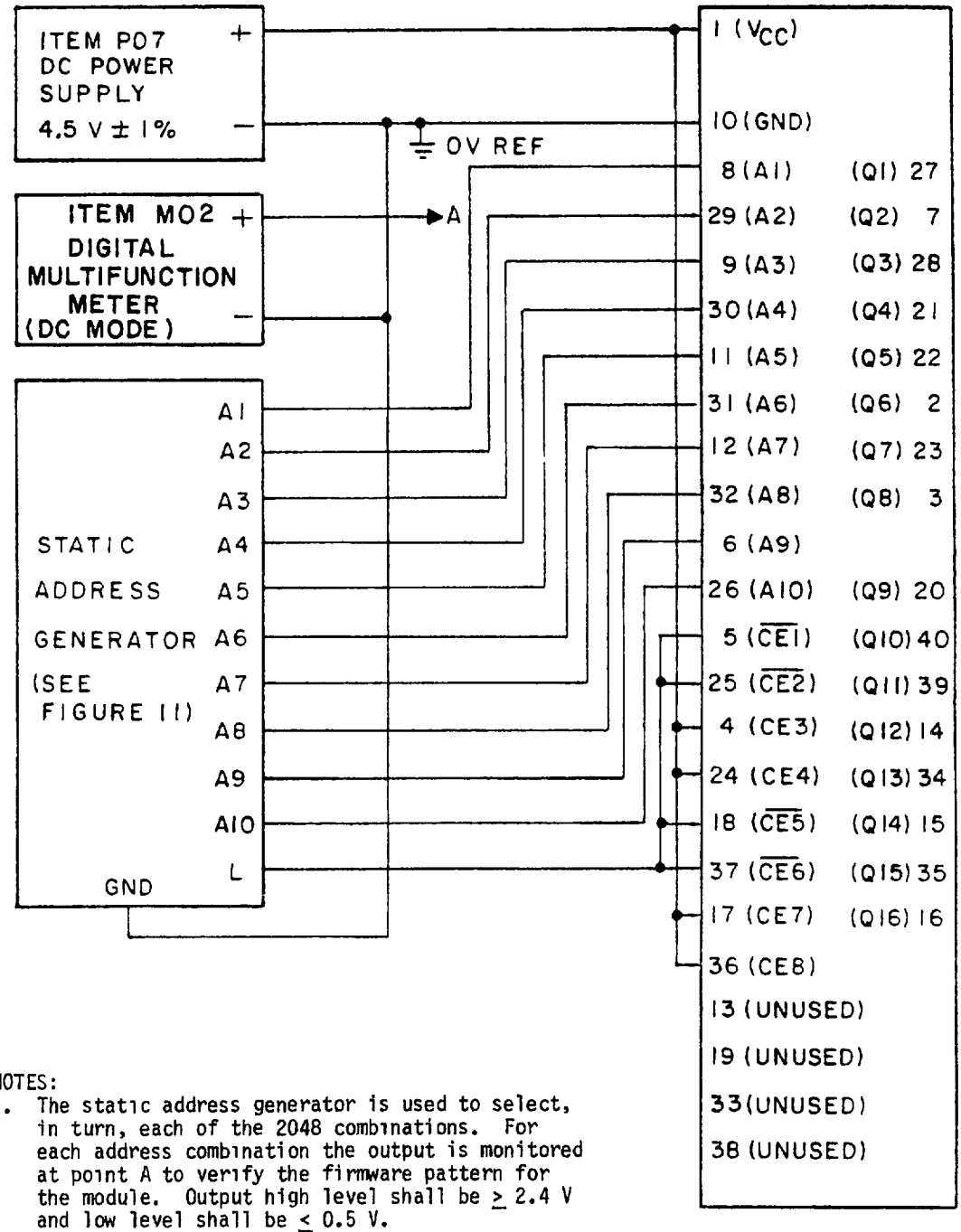
TABLE XVIII. Functional test.

Test condition	Parameter	Initial limits 25°C ±5°C ambient		Unit	Test figure
		Min	Max		
S	Functional	See note 1, figure 27			27

TABLE XIX. Power supply transient amplitude test.

Test condition	Parameter	Initial limits 25°C ±5°C ambient		Unit	Test figure
		Min	Max		
T	Power supply transient amplitude	0	400	mV	28,29

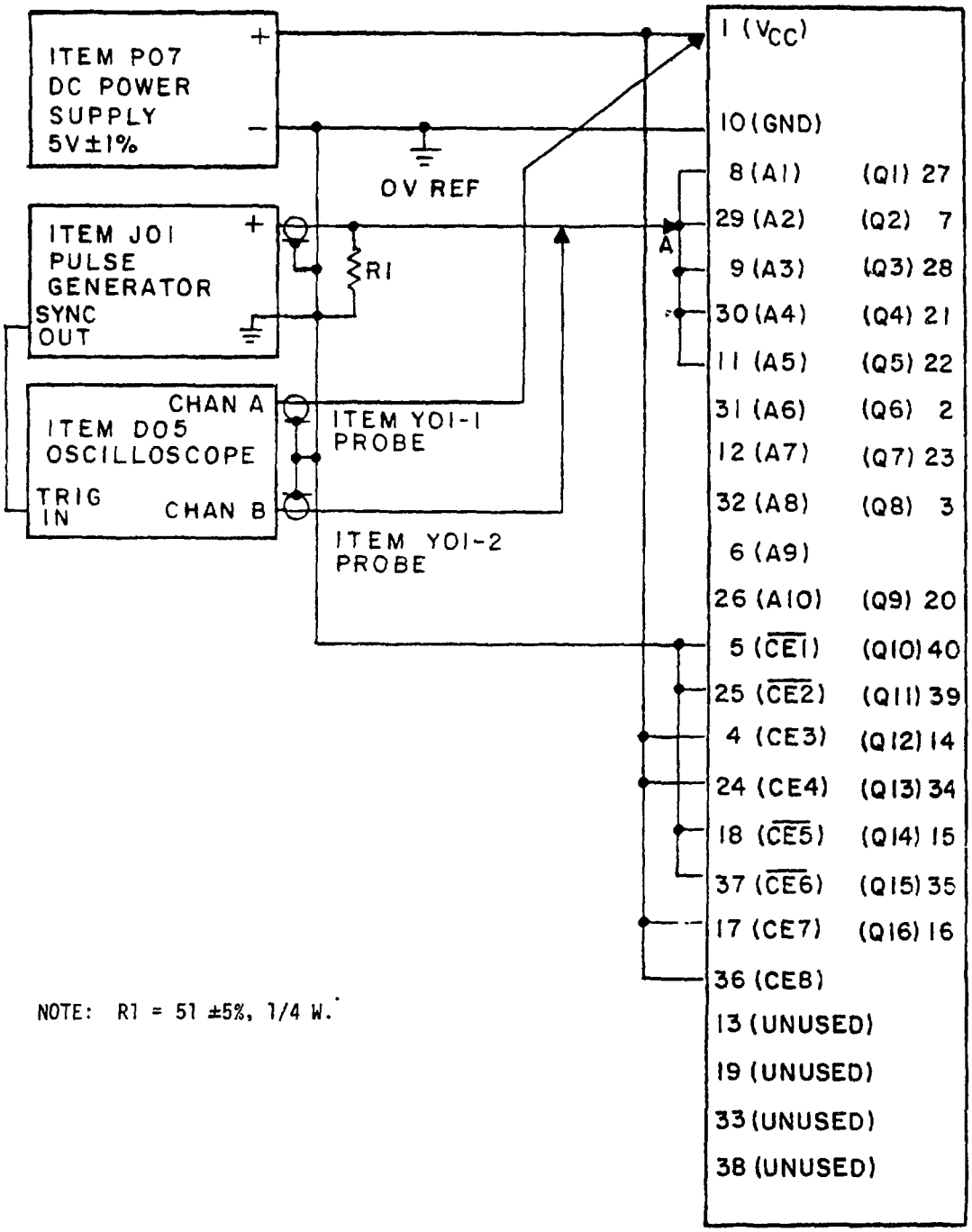
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NOTES:
1. The static address generator is used to select, in turn, each of the 2048 combinations. For each address combination the output is monitored at point A to verify the firmware pattern for the module. Output high level shall be ≥ 2.4 V and low level shall be ≤ 0.5 V.
2. Load the otuput under test with the load as specified on figure 16.

FIGURE 27. Functional test circuit for table XVIII.

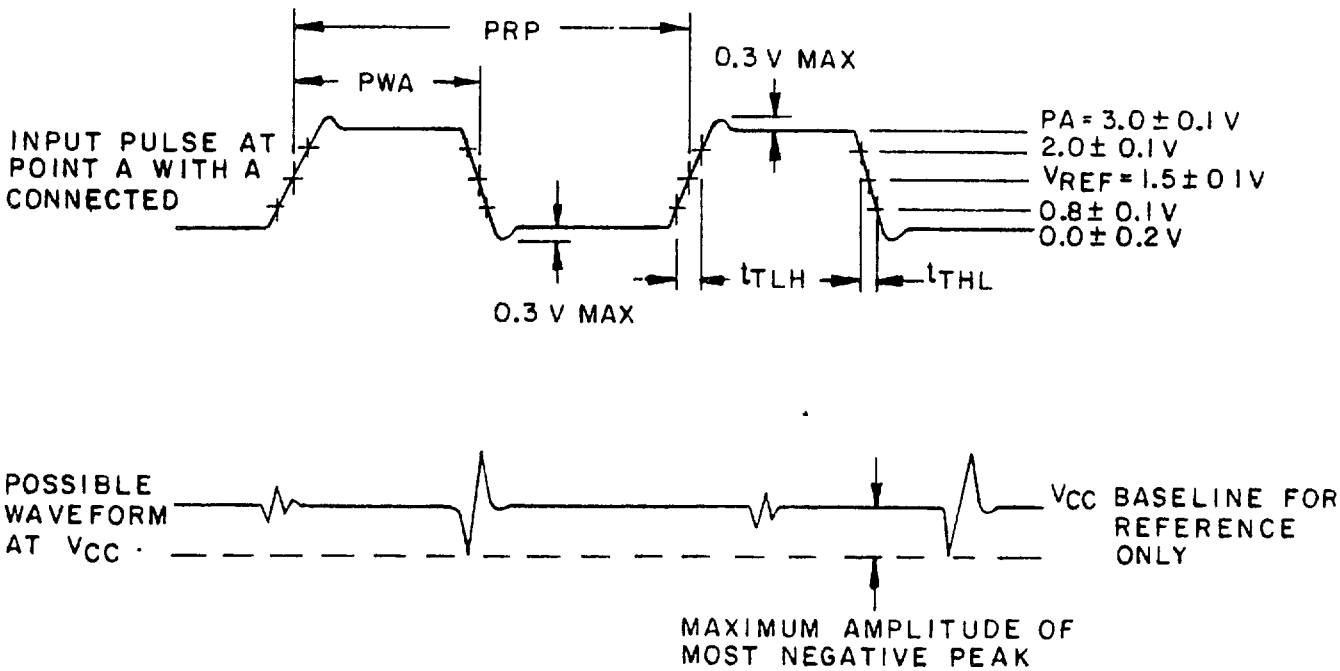
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NOTE: R1 = 51 ±5%, 1/4 W.

FIGURE 28. Power supply transient amplitude test circuit for table XIX.

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NOTES:
1. Input pulse parameters:
   $t_{TLH} = t_{THL} = 3.5 \pm 1.5$ ns (verified at each input pin).
   PWA = 1.0 ±0.1 µs.
   PRP = 2.0 ±0.1 µs.
2. All voltages are referenced to ground.
3. Transient amplitude is measured from the steady state V_{CC} level to the most negative excursion of the waveform.

FIGURE 29. Power supply transient amplitude test circuit waveforms for table XIX.

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TABLE XX. Power supply filter capacitance test.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
U	Power supply filter capacitance	Open	1	1.6	3.0	µF	30

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TABLE XXI. Input capacitance test.

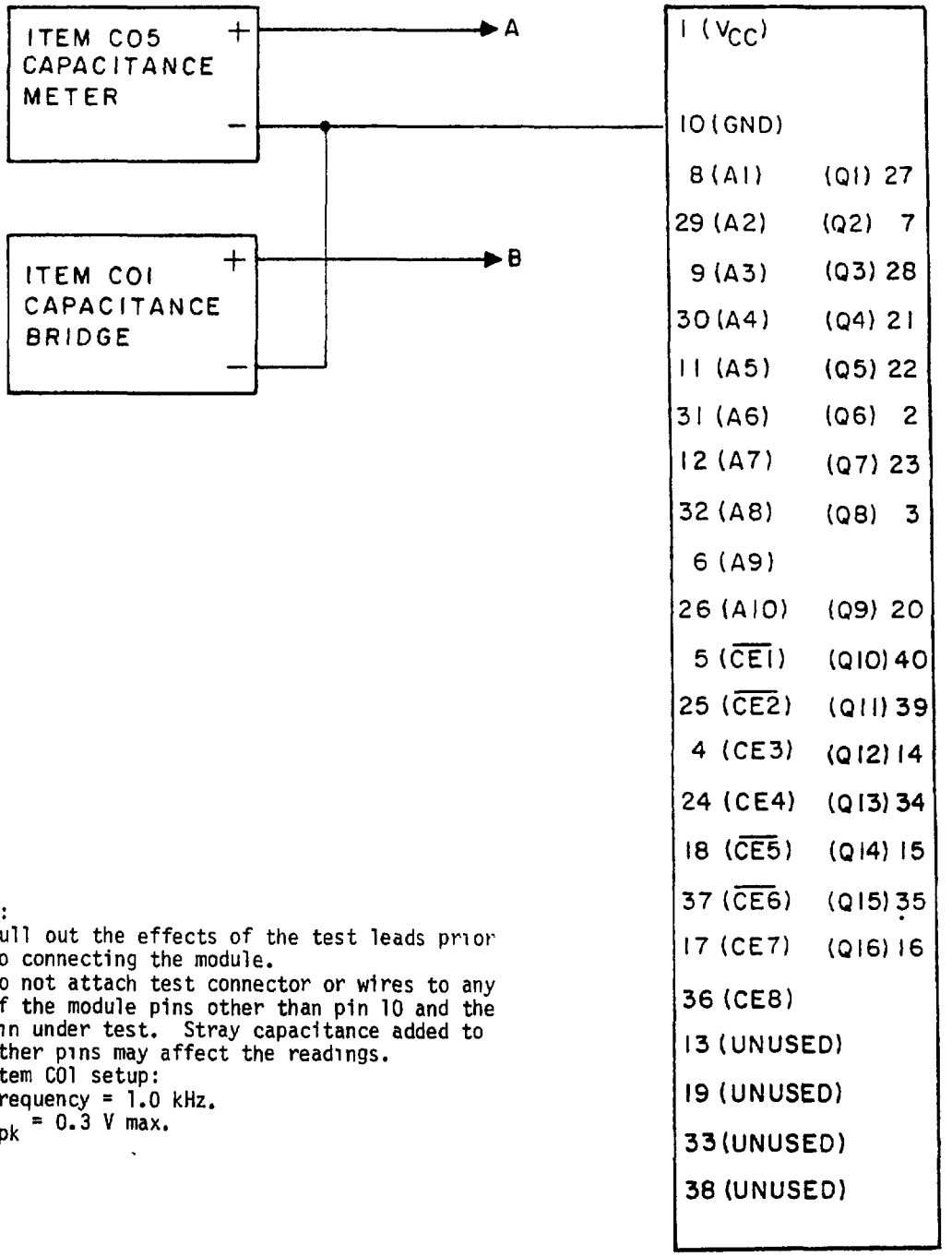
Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
V1	A1 Input capacitance	8	Open	2	30	pF	30
V2	A2 "	29	"	"	"	"	"
V3	A3 "	9	"	"	"	"	"
V4	A4 "	30	"	"	"	"	"
V5	A5 "	11	"	"	"	"	"
V6	A6 "	31	"	"	"	"	"
V7	A7 "	12	"	"	"	"	"
V8	A8 "	32	"	"	"	"	"
V9	A9 "	6	"	"	"	"	"
V10	A10 "	26	"	"	"	"	"
V11	$\overline{CE1}$ "	5	"	"	15	"	"
V12	$\overline{CE2}$ "	25	"	"	"	"	"
V13	CE3 "	4	"	"	"	"	"
V14	CE4 "	24	"	"	"	"	"
V15	$\overline{CE5}$ "	18	"	"	"	"	"
V16	$\overline{CE6}$ "	37	"	"	"	"	"
V17	CE7 "	17	"	"	"	"	"
V18	CE8 "	36	"	"	"	"	"

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TABLE XXII. Output capacitance test.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
W1	Q1 Output capacitance	27	Open	2	15	pF	30
W2	Q2 "	7	"	"	"	"	"
W3	Q3 "	28	"	"	"	"	"
W4	Q4 "	21	"	"	"	"	"
W5	Q5 "	22	"	"	"	"	"
W6	Q6 "	2	"	"	"	"	"
W7	Q7 "	23	"	"	"	"	"
W8	Q8 "	3	"	"	"	"	"
W9	Q9 "	20	"	"	"	"	"
W10	Q10 "	40	"	"	"	"	"
W11	Q11 "	39	"	"	"	"	"
W12	Q12 "	14	"	"	"	"	"
W13	Q13 "	34	"	"	"	"	"
W14	Q14 "	15	"	"	"	"	"
W15	Q15 "	35	"	"	"	"	"
W16	Q16 "	16	"	"	"	"	"

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NOTES:
1. Null out the effects of the test leads prior to connecting the module.
2. Do not attach test connector or wires to any of the module pins other than pin 10 and the pin under test. Stray capacitance added to other pins may affect the readings.
3. Item C01 setup:
   Frequency = 1.0 kHz.
   V_{pk} = 0.3 V max.

FIGURE 30. Power supply filter, input, and output capacitance test circuit for tables XX, XXI, and XXII.

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